



DEDICATION

This work is dedicated to the Forche's Family for their
countless encouragement and support throughout the
academic year.

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ABBREVIATIONS

- **2TUP:** Two Track Unified Process
- **AICS:** African Institute of Computer Sciences
- **API:** Application Programming Interface
- **CSS:** Cascading Style Sheet
- **DBMS:** Database Management System
- **DIY:** Do It Yourself
- **GUI:** Graphical User Interface
- **HTML:** Hypertext Markup Language
- **HTTP:** Hypertext Transfer Protocol
- **IDE:** Integrated Development environment
- **JSON:** JavaScript Object Notation
- **KYC:** Know Your Customer
- **OS:** Operating System
- **UML:** Unified Modelling Language

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GLOSSARY

- 2TUP: Two Track Unified Process.
- AICS: African Institute of Computer Sciences.
- DBMS: Database Management System
- MVC: Model View Controller
- UML: Unified Modelling Language.

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ABSTRACT

This project, titled “**Conception and Realization of an Employee Attendance and Payroll Management System,**” was developed during our internship at **REALIZE** with the objective of addressing the limitations of manual attendance and payroll processes, which are often time-consuming, error-prone, and inefficient. The system provides a centralized digital platform that automates attendance tracking, leave management, justification processing, and payroll generation, while integrating features such as location-based login, role-based access, and automated timesheets to ensure efficiency, transparency, and accountability. Using **UML modeling** for system design and a modern technological stack (React/Bootstrap for the interface, Node.js/Django/Laravel for the backend, and MySQL/PostgreSQL for data storage), the platform is designed to be scalable, secure, and user-friendly. The project not only improves workforce management and reduces administrative workload but also lays the foundation for future extensions such as QR/NFC-based login, AI-powered analytics, and mobile app integration, thereby enhancing organizational efficiency and supporting digital transformation in employee management.

Keywords: Employee Management, Payroll System, Attendance Tracking, Leave Management, Digital Transformation, UML, Web Application, Workforce Automation

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RESUME

Ce projet, intitulé « Conception et Réalisation d'un Système de Gestion des Présences et de la Paie des Employés », a été développé dans le cadre de notre stage chez REALIZE avec pour objectif de répondre aux limites des processus manuels de suivi des présences et de gestion de la paie, souvent chronophages, sujets aux erreurs et inefficaces. Le système fournit une plateforme numérique centralisée qui automatise le suivi des présences, la gestion des congés, le traitement des justifications et la génération de la paie, tout en intégrant des fonctionnalités telles que la connexion basée sur la localisation, l'accès basé sur les rôles et la gestion automatisée des feuilles de temps afin de garantir efficacité, transparence et traçabilité. En utilisant la modélisation UML pour la conception du système et un socle technologique moderne (React pour l'interface, Node.js express pour le backend et Mongo DB pour le stockage des données), la plateforme est conçue pour être évolutive, sécurisée et conviviale. Ce projet permet non seulement d'améliorer la gestion des employés et de réduire la charge administrative, mais il pose également les bases d'évolutions futures telles que la connexion par QR Code/NFC, l'analyse prédictive basée sur l'IA et l'intégration mobile, renforçant ainsi l'efficacité organisationnelle et soutenant la transformation numérique de la gestion des ressources humaines.

Mots-clés : Gestion des employés, Système de paie, Suivi des présences, Gestion des congés, Transformation numérique, UML, Application web, Automatisation des ressources humaines.

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GENERAL INTRODUCTION

In today's rapidly evolving business environment, efficient workforce management has become a cornerstone for organizational success. Companies across various industries increasingly face challenges in monitoring employee attendance, managing leave requests, processing payroll, and ensuring accountability in daily operations. Traditional manual systems or fragmented digital tools often lead to inaccuracies, delays, lack of transparency, and administrative inefficiencies, which can negatively impact productivity and employee satisfaction.

The **Realize Management System** project aims to overcome these limitations by developing a modern, secure, and user-friendly platform that automates key aspects of workforce management. By integrating functionalities such as real-time attendance tracking, geolocation-based login, leave and justification management, and automated payroll processing, the system seeks to provide organizations with a reliable and efficient solution tailored to their operational needs.

This platform is designed to support a variety of users, including employees, supervisors, and administrators, each with clearly defined roles and responsibilities. The system is built using the **MVC (Model-View-Controller) architecture**, with a responsive **React and Tailwind CSS** frontend, and a secure **Node.js, Express, and MongoDB** backend to ensure scalability, accuracy, and security.

Through this project, the objective is to improve organizational efficiency, reduce administrative burdens, enhance transparency, and deliver a seamless digital experience for workforce management. The **Realize Management System** represents a significant step forward in leveraging technology to optimize human resource processes, enabling organizations to focus more on growth and innovation.

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PART I: INSERTION PHASE

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PREAMBLE

The inserted document forms part of my internship report, which details the structure and objectives of the organization where I completed my internship. It presents an overview of my internship environment and delineates the different stages I underwent to successfully assimilate into the company.

CONTENT

INTRODUCTION

I. WELCOMING AND INTEGRATION

II. GENERAL PRESENTATION OF REALIZE

III. ORAGANIZATION OF REALIZE

IV. HARDWARE AND SOFTWARE RESOURCES

V. BREIF PRESENTATION OF PROJECT

CONCLUSION

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INTRODUCTION

The insertion phase marked my first glimpse into both professional and general realities when we began working as an intern. Throughout this phase, we familiarized ourselves with the reception structure, comprehended its operation, researched administrative necessities to better understand the predicaments facing us, and ultimately resolved an issue that was compliant with the structure's demands. Given that I had already interned at REALIZE last year, this phase was smoother for me. I was already acquainted with the organization's environment and processes, so it was more about catching up on the new developments and refining my understanding. The proposed insertion phase report details our initial foray into the professional world, particularly within REALIZE. It pertains to the period between the 8th and 19th of July 2024, focusing on REALIZE and its structure, along with the introduction of my host unit—the Software Engineering department—and my seamless assimilation into the internship program there.

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I. WELCOMING AND INTEGRATION

The welcoming and integration event took place on the 8th of July at 8:30 am. Upon arrival, we were greeted by Mr. Mengot, the senior developer, who delivered an inspiring welcoming speech. Having interned at REALIZE last year, this moment felt familiar, though it was refreshing to reconnect with the team. We proceeded with a presentation and a meeting focused on discussing the challenges we faced during the coding test, which was a prerequisite for our internship at the company. The meeting also included an overview of the enterprise's structure, its internal policies, and the specific workstations assigned to each intern. Given my previous experience, I was able to quickly reorient myself within the company's environment. Mr. Tanue concluded by encouraging us to strive for excellence and offered his support for any issues we might encounter.

After settling into our designated workstations, we were promptly assigned our first task for the day. During the first week, we conducted a verification of concepts using fundamental HTML and CSS to ensure everyone was on an even playing field before progressing. We then engaged in discussions about project ideas, where we were encouraged to propose our own. My previous experience made it easier to contribute and refine these ideas. Subsequently, we were tasked with researching these ideas, as the enterprise valued our initiative and did not impose any themes on us.

In the second week, we focused on JavaScript concept verification exercises to assess our understanding and reasoning skills. We held a detailed discussion with our professional supervisor about the specification book and its various components, where he outlined what was expected of us in each segment. We also continued developing our project ideas into a concrete plan, benefiting from my previous familiarity with the company's expectations and processes.

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II. GENERAL PRESENTATION OF REALIZE

i. Background

Realize is a Cameroon based non-governmental tech start-up institute founded by NDELOKAKEH Daniel in 2022 which proposes IT solutions and empowers the use of new technologies in Cameroon.

ii. Missions, Visions and Activities of Realize

1. Missions:

The mission of Realize mainly relates to the empowerment and the perpetuation of Technology use. These missions include:

- Design and realize mobile-oriented software for companies and individuals;
- Assist newborn companies for quick growth using computer sciences;
- Provide training and certifications to improve qualified human resources in many fields of study;
- Take an active part in the sustainable development of the world through innovative solutions and virtual reality.

2. Vision:

At Realize, we believe that the true potential of every organization and individual can be unlocked through the harmonious fusion of analysis, development, realization, and data analytic. Our vision propels us forward, guiding us to make a lasting impact on the digital landscape and shape a future where innovation knows no bounds."



3. Activities:

The activities of Realize range from computer sciences, engineering, and training. We can outline the following:

- Conception realization, and hosting of websites;
- Software development and maintenance;
- Training in Software related fields;
- Conception and realization of multimedia;
- IT consulting and innovation
- IT support



Localization



Figure 1: LOCALIZATION PLAN OF REALIZE

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III.ORGANIZATION OF REALIZE

A. Administrative organization

Realize if administratively organized as follows:

a. The General management

This is the highest level of the company, which takes care of the following:

- Proper functioning of each department;
- Define project strategies;
- Provide leadership and guidance to the company's employees.
- Makes critical decision that could affect the company's operations or reputation
- Severe as a point of contact for key stakeholders such as investors etc.

b. Human resource department

This department is in charge of the following:

- Recruitment and Hiring
- Acts as a liaison between employees and management;
- Manages employee compensation and benefits programs;
- Develops and implements company policies and procedures, ensuring compliance with employment laws and regulations.
- Manages administrative tasks related to employee records, data management, and HR systems.



c. Communication Department

This department is in charge of the following:

- Handles public relations activities, which involve managing the company's reputation and image;
- Responsible for crafting and delivering messages to external stakeholders such as customers etc.
- It fosters effective communication within the company;
- They create and manages content across different platforms and channels;

d. Department of Financial Affairs

This department is in charge of the following:

- Responsible for developing and managing the company's financial plans and budgets;
- Prepare and presents accurate and timely financial reports to management, stakeholders, and regulatory authorities;
- Responsible for managing cost and expenses within the organization;
- Ensures compliance with tax laws and regulations. Etc.



e. Technical department

This department is in charge of the following:

- Responsible for managing the company's technological infrastructure, including networks, hardware;
- Play a role in managing and maintaining the company's data;
- Responsible for implementing and maintaining cybersecurity measures to protect the company's digital assets from potential threats;
- Interacts with technology vendors and manages relationships with external service providers. Etc.

f. Software Engineering Department

This department is in charge of the following:

- Primarily responsible for developing software applications and systems;
- Responsible for maintaining and supporting software application throughout their Life cycle;
- Evaluation and realization of projects etc.



B. FUNCTIONAL ORGANIZATION OF REALIZE

The functional branch of Realize is organized as follows

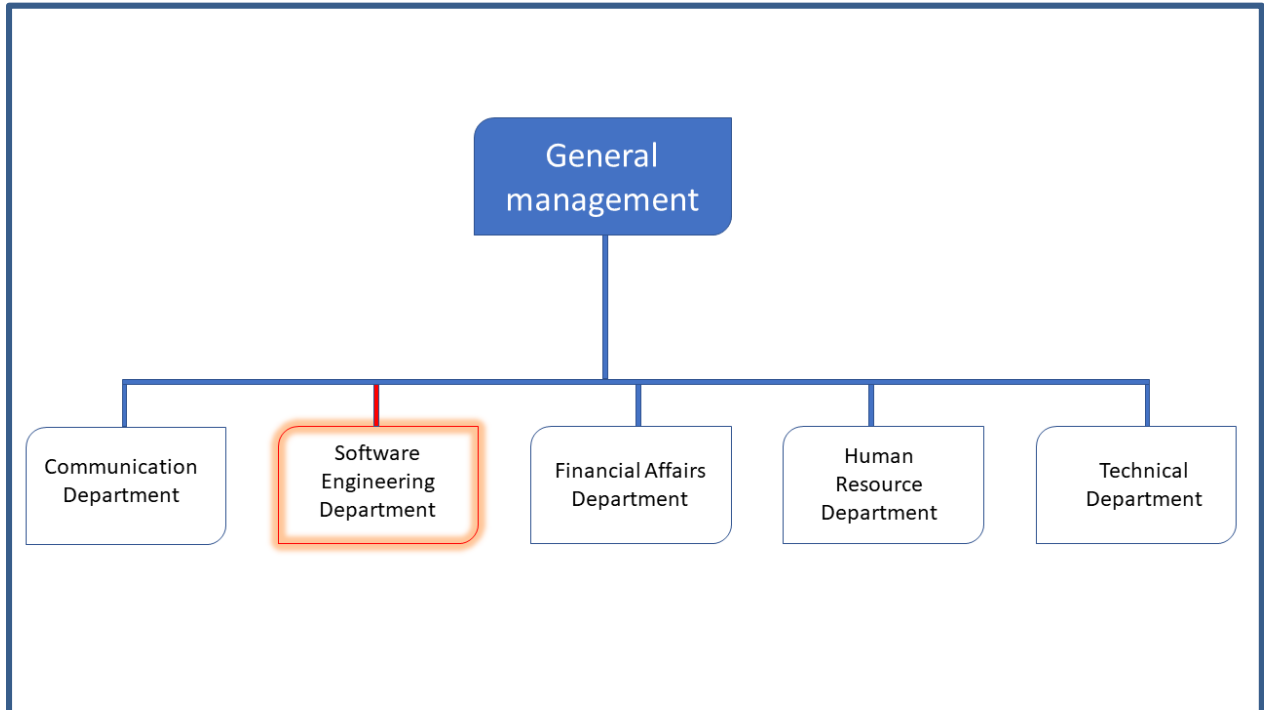


Figure 2: FUNCTIONAL ORGANIZATION OF REALIZE



IV. HARDWARE AND SOFTWARE RESOURCES

A. HARDWARE RESOURCES

Table 1: HARDWARE RESOURCES

No	DESIGNATION	QUANTITY	CHARACTERISTICS	OBSERVATION
1	Mac Book pro	1	APPLE	Good
2	Modem	1	CAMTEL	Good
3	LCD screen	1	HP	Excellent
4	Training equipment and office furniture	/	/	

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B. SOFTWARE RESOURCES

Table 2: SOFTWARE RESOURCES

DESIGNATION	SOFTWARE
Operating System	Windows 10, Mac OS, parrot Linux
Design tools	Photoshop
Integrated development environment (IDE)	Visual studio code
Text editor	Sublime text, notepad++
Database management system (DBMS)	MongoDB, PostgreSQL, MySQL
Web browser	Google chrome, Microsoft edge
Document editor	Microsoft office word

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V. BRIEF PRESENTATION OF PROJECT

At the end of the initial two weeks (insertion period), we received a topic, namely **DESIGN AND IMPLEMENTATION OF A WEB APPLICATION FOR EMPLOYEE ATTENDANCE TRACKING AND PAYROLL MANAGEMENT**. This project involves developing a platform designed to monitor employees' daily attendance, manage check-in/check-out times, and calculate their total working hours automatically. The system also integrates payroll, ensuring that employees are paid according to the exact time spent on-site.

In addition to attendance, the application supports leave management, where employees can submit leave requests through their superiors, with approvals directly impacting payroll. Employees can also justify lateness or absence, ensuring accountability and transparency. The primary goal of the application is to simplify workforce management, reduce payroll errors, and provide a seamless experience for employees, supervisors, and HR administrators.

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CONCLUSION

Our integration into the company was a pleasurable experience, with friendly and approachable colleagues and a professional supervisor who made us feel welcomed. Their availability and smooth integration into the company's operations were commendable. One of the most valuable takeaways from this phase was the importance of collaboration and discipline, crucial to any successful career. The subsequent phase of this momentous journey involves the project specifications, which must be precise and succinct to comprehensively understand what is needed to extract the most value from this transformative project.

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PART II: TECHNICAL PHASE

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FILE 1: EXISTING SYSTEM

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PREAMBLE

This chapter will present and elucidate our central theme, which is the primary emphasis and objective of our project. We shall also expound on the predicament that prompted us to devise our proposed solution, and detail the characteristics and operations of the software we have planned to construct.

CONTENT

INTRODUCTION

I. THEME PRESENTATION

II. STUDY OF THE EXISTING SYSTEM

III. LIMITATION OF THE EXISTING SYSTEM

IV. PROBLEMATIC

V. PROPOSED SOLUTION

CONCLUSION

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INTRODUCTION

In this part of our report, we will explain our theme. We will also study the current system and identify its strengths and weaknesses. Based on this analysis, we will present the problem statement, which is the main challenge or gap that the current system faces. Finally, we will propose a solution that addresses this problem and improves the system.

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THEME PRESETATION

The theme of this project is the **Conception and Realization of a Web Application for Employee Attendance Tracking and Payroll Management**, designed to streamline workforce management by accurately monitoring employee presence and calculating payroll based on actual hours worked. The application ensures that employees can check in and check out only when physically present at the work site, thereby maintaining precise attendance records and promoting accountability.

Realize provides a comprehensive platform where employees can submit leave requests through their superiors, justify lateness or absence, and view their total work hours. Supervisors and HR administrators can monitor team attendance, approve leaves and justifications, and generate payroll reports efficiently. The application is designed with flexibility and scalability in mind, aiming to provide a secure, transparent, and user-friendly system that reduces administrative workload, minimizes payroll errors, and enhances overall workforce management.

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III. LIMITATIONS OF THE EXITING SYSTEM

LIMITATION	CONSEQUENCES	PROPOSED SOLUTION
Attendance is recorded manually (registers, spreadsheets)	Errors, falsification, loss of record	Implement automated attendance system with GPS/IP verification
Payroll is calculated manually	Salary inaccuracies (underpayment or overpayment)	Link attendance records to an integrated payroll system
Leave request and justifications are verbal or via email	Delays, miscommunication, and poor record keeping	Introduce a digital workflow for leave requests and justifications with superior's approval
No real time monitoring of attendance	HR and supervisor cannot detect absenteeism or lateness instantly	Provide real time dashboards for supervisors and HR
No location verifications for check in	Employee can mark attendance without being present on site	Enforce onsite check-ins using GPS or IP based validation

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**DESIGN AND IMPLEMENTATION OF A WEB APPLICATION FOR
EMPLOYEE ATTENDANCE TRACKING AND PAYROLL MANAGEMENT (case
study realize enterprise)**



Processes are fragmented (attendance, leave, payroll, handled, separately)	Increased administrative workload, inefficiency	Develop a unified and scalable web platform that centralizes all process

Table 3:limitations



IV. PROBLEMATIQUE

The current methods for managing employee attendance and payroll are inefficient, error-prone, and lack transparency. Manual processes or partially digital systems fail to provide real-time monitoring, accurate work hour calculations, or a seamless leave approval workflow. This results in:

- Payroll discrepancies and delayed payments.
- Time-consuming administrative tasks for HR and supervisors.
- Lack of accountability and traceability for employees' attendance and leave.
- Reduced employee satisfaction due to unclear records or approval delays.

There is a clear need for a unified, automated system that ensures accuracy, accountability, and timely payroll processing while simplifying the workflow for both employees and management.



V. PROPOSED SOLUTION

The **Realize web application** addresses the limitations of the existing system by providing:

1. **Automated Attendance Tracking:** Employees check in and out only on-site, with GPS or IP-based verification. Total work hours are calculated automatically.
2. **Integrated Payroll Management:** Payroll is generated based on actual hours worked and approved leaves, reducing errors and administrative burden.
3. **Leave and Justification Workflow:** Employees submit leave requests or lateness/absence justifications through the app. Supervisors approve or reject requests directly, ensuring transparency.
4. **Real-Time Monitoring:** Supervisors and HR can track team attendance, leave requests, and payroll reports in real-time.
5. **Unified Platform:** Attendance, leave management, and payroll are integrated into a single system, improving efficiency and accountability.
6. **Secure and User-Friendly:** The app is accessible via web browsers, ensuring secure login and an intuitive interface for all user types.



CONCLUSION

The Realize project aims to modernize employee attendance and payroll management by replacing error-prone manual systems with a secure, automated, and transparent web application. By integrating attendance tracking, leave management, and payroll calculation, the platform ensures accurate work hour monitoring, timely salary payments, and streamlined administrative workflows.

Realize not only improves operational efficiency for HR and supervisors but also enhances accountability and transparency for employees. This project demonstrates how technology can simplify workforce management, reduce errors, and create a more reliable and fair system for both employers and employees.

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FILE 2: SPECIFICATION BOOK

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PREAMBLE

The specification book serves as a contractual document before the application development process commences. It succinctly outlines the expectations of the project owner and the respective responsibilities of all parties involved. It provides a specific and detailed description of the topic and user requirements, as well as the necessary conditions for project execution. The project's success will depend on careful allocation and management of the various resources available, including physical, financial, and human assets. Therefore, this document outlines the criteria that the developer must adhere to in order to ensure the success of the project.

CONTENT

INTRODUCTION

I. CONTEXT AND JUSTIFICATION

II. OBJECTIVES OF THE PROJECT

III. EXPRESSION OF NEEDS

IV. PROJECT PLANNING

V. ESTIMATED COST OF THE PROJECT

VI. PROJECT CONSTRAINTS

VII. LIST OF PARTICIPANTS AND
DELIVERABLES

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INTRODUCTION

The report's specification book enables us to provide comprehensive details about our topic, enhancing our understanding and improving its chances of success. To establish the project's scope, we will describe the context of our topic and identify the issues within it that we aim to solve. Once the solution is proposed, we will outline the objectives we have set for ourselves and for the project. Also, we will assess the functional and non-functional requirements of our system before designing a plan to ensure timely completion of the project and determine the financial resources needed. Subsequently, we will outline the expected project deliverable at the conclusion of the project.

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CONTEXT AND JUSTIFICATION

CONTEXT:

In modern organizations, accurate tracking of employee attendance and timely payroll management are essential for operational efficiency and workforce accountability. Many companies, especially those with multiple sites or remote operations, face challenges with traditional manual attendance systems, such as paper registers or spreadsheets. These methods are prone to errors, time-consuming, and lack transparency, which can lead to payroll discrepancies, delays in salary payments, and employee dissatisfaction.

Furthermore, leave management and absence justifications are often handled through emails or verbal approvals, resulting in miscommunication, delayed decisions, and incomplete records. Supervisors and HR personnel spend significant time manually verifying attendance, calculating work hours, and reconciling payroll, reducing their focus on strategic tasks.

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2. JUSTIFICATION

The Realize web application is justified by the need to:

1. **Improve Accuracy:** Automate attendance tracking and payroll calculation to minimize errors and ensure employees are compensated fairly.
2. **Enhance Efficiency:** Reduce administrative workload by centralizing attendance, leave requests, justifications, and payroll into a single platform.
3. **Ensure Accountability:** Enable supervisors to monitor attendance, approve leaves, and track justifications in real-time, creating a transparent work environment.
4. **Support Decision-Making:** Provide HR and management with accurate reports and analytics to make informed workforce management decisions.
5. **Adapt to Modern Work Practices:** Incorporate site-based login verification and digital workflows that align with current technological standards and employee expectations.

By addressing the limitations of existing manual or fragmented systems, Realize offers a secure, scalable, and user-friendly solution that benefits employees, supervisors, and HR departments alike. It ensures that workforce management is accurate, transparent, and efficient, while providing a reliable foundation for organizational growth and operational excellence.



OBJECTIVES OF THE PROJECT

1. General Objective

The general objective of this project is to design and implement a web based system that automates employee attendance tracking, leave management, and payroll computation in order to improve accuracy, transparency, and efficiency in workforce management

2. Specific Objectives

To achieve the general objective, the project aims to

- ❖ Automate attendance tracking
By enabling employees to check in and out only when physically present on site
- ❖ Accurately calculate work hours and payroll: by linking attendance records directly with the salary computation
- ❖ Digitize leave management: by allowing employees to request leave or justify absences through the system, with supervisors able to approve or reject requests
- ❖ Provide Real-Time monitoring tools: for supervisors and HR to track attendance, leave, status, and payroll in one platform.
- ❖ Enhance accountability and transparency: by giving employees access to their attendance history, payroll records, and superior hierarchy.
- ❖ Reduce administrative workload: by centralizing attendance, payroll, and leave management in one unified platform.
- ❖ Ensure scalability and flexibility: so the system can handle multiple sites departments, and varying pay structure.

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The project is characterized by the following:

- ❖ **Project Name:** REALIZE HR system
- ❖ **Project Target:** enterprises and businesses seeking reliable and efficient attendance tracking and payroll
- ❖ **Technical Specification:** Web application (React JSX) integrated with a backend (Node.js express, MongoDB) for real-time data management, authentication, and payment processing.

III. EXPRESSION OF NEEDS

1. FUNCTIONAL NEEDS

In this context, functional requirements describe what the: REALIZE HR system or application should do. The modules are as follows:

Customer:

User Authentication: secure login for employees, supervisors, and HR administrators.

Attendance tracking:

- Employees check in and check out
- System records arrival and departure time
- Automatic calculation of total work hours

Location verification: attendance can only be marked if the employee is physically on present on site (via GPS/IP).



Payroll management:

- System calculates salaries based on hours worked
- Approved leave days are integrated as paid time
- Payroll reports generate for HR

Leave management

- Employees request leave through the system
- Supervisors approve or reject requests
- HR monitors leave balances

Absence/Justification management: employees can justify lateness or absences; supervisor validated or reject.

Organizational Hierarchy:

- Employees can see their direct superior
- Supervisors can see attendance/leave status of their team

Reporting and Monitoring:

- HR and supervisors can generate daily, weekly, or monthly attendance/payroll reports
- Exportable in PDF/Excel



2. NON-FUNCTIONAL NEEDS

In this section, non-functional requirements define the quality attributes, standards, and performance measures that the REALIZE HR system must adhere to. These aspects ensure the overall performance, security, compliance, and usability of the application.

1. Security:

- Role based access (Employee, Superior, HR/Admin)
- Data encryption and secure authentication

2. Performance: the system should process attendance and payrolls quickly without delays.

3. Scalability Requirements: simple, user friendly dashboards for employees and administrators.

4. Reliability and Availability: system must ensure accurate and consistent time tracking without data loss, high system uptime to ensure employee can check in/out

5. Scalability: able to handle multiple branches, departments, and increasing numbers of employees

6. Auditability: every check-in, leave request, or payroll update is logged for accountability

7. Maintainability: easy to update and adapt to organizational policy changes

8. Compatibility: accessible via web browsers and adaptable to mobile devices.



IV. PROJECT PLANNING

a. Chronogram of activities

PHASE	OBJECTIVE	OUTPUT	DURATION	PERIOD
INSERTION	Collection of information on the enterprise	Insertion book	2 weeks	01th July to 18th July
EXISTING SYSTEM	Study of the existing system	Existing System	3days	21 st July to 23 rd July
SPECIFICATION BOOK	Specification of the user needs	Specification book	9days	24 th July to 5 th August
ANALYSIS	Capture of needs Use case and textual description Modelling	Analysis Book	5days	06 th August to 12 th August
CONCEPTION	Preliminary conception and Detailed conception	Conception book	13days	13 ^t August to 29 th August
REALIZATION	Implementation Unitary Test Integration Development	Realization book	12days	01 st September To 16 th September Development

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TEST OF FUNCTIONALITIES	Testing of the software and debugging	Test Of Functionalities	5days	17 th September to 123 th September
INSTALLATION AND USER GUIDE	Documenting software	User Guide	5days	24 nd September to 30 th October

Table 4: Chronogram of activities

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b. Gantt Project Planning

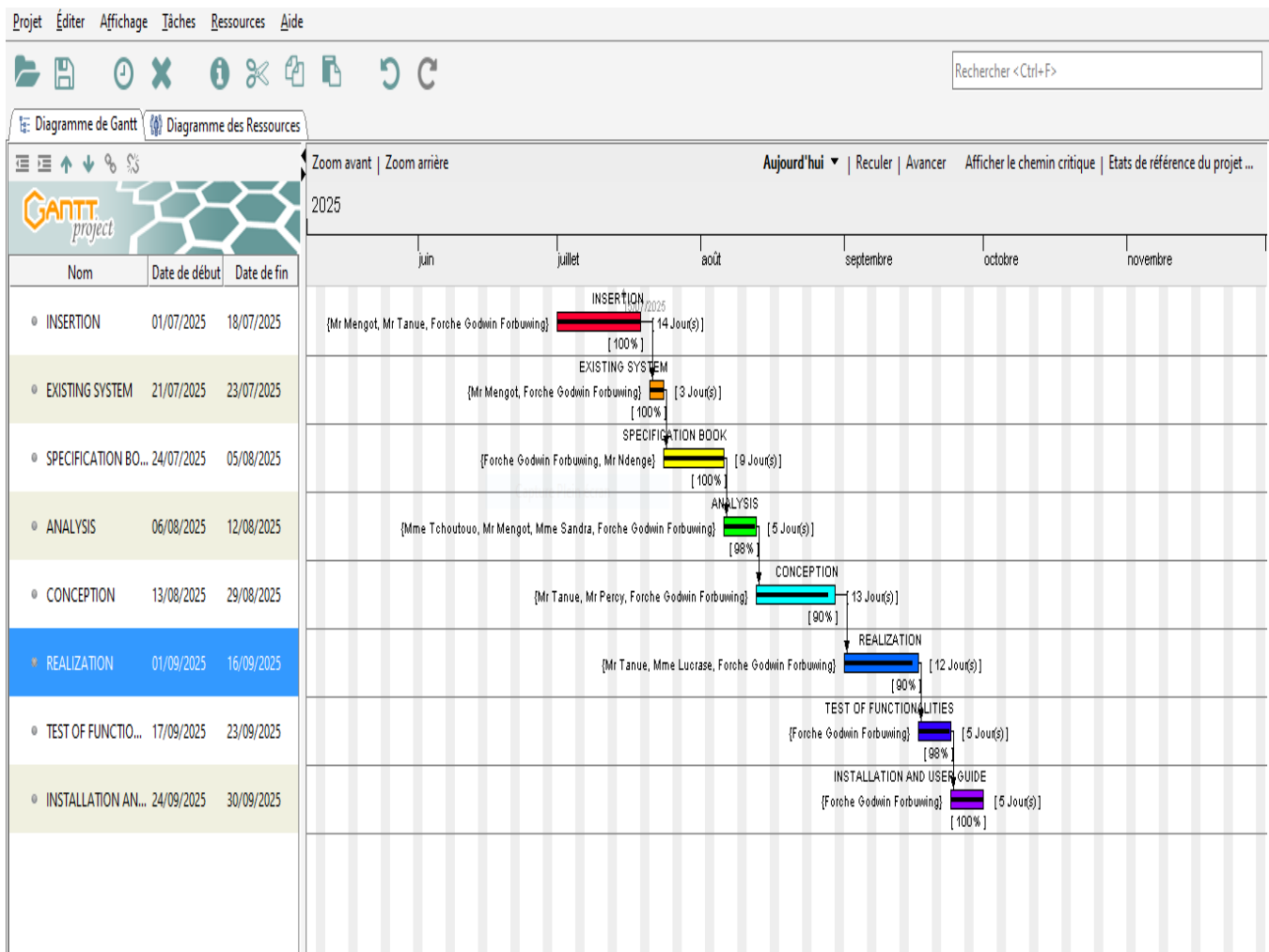


Figure 3:PROJECT GANTT CHART

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V. ESTIMATED COST OF THE PROJECT

a. Software Resources

Table 5: SOFTWARE RESOURCES (SOURCE: MERCURIALE-2025)

RESOURCES	DESIGNATION	USAGE	QUANTITY	UNIT COST (FCFA)
Formatting	MICROSOFT OFFICE PROFESSIONAL 2019	Writing of the report	1	355465
Web browser	Google Chrome	Accessing and view web pages on the internet	1	Freeware
Code Editor	Visual Studio Code	For writing the code of the application	1	Freemium
Project Planning	Gannt Project	For building a Gantt chart	1	Freemium
Graphic Editor	Icogram	For building the geographical location of the enterprise	1	Freemium
UML Analysis	visual paradigm	For drawing visual paradigms	1	Community Version

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		diagrams		
Version Control	Git & GitHub	For tracking and managing changes to software code	1	Free Version
TOTAL 1		7		355465

b. Hardware Resources

Table 6: *HARDWARE RESOURCES* (SOURCE: MERCURIALE-2025)

RESOURCES	HARDWARE	QUANTITY	UNIT (FCFA)	COST
Computer	DELL LATITUDE 5490 CORE I5 vPro 7th Gen, HARD DISK 500 SSD, RAM 16GB	1	1295000	
Printer	Printers	1	514528	
Network	Local network installation	1	300 000	
Removable Disk	USB HP 2.2 FLASH DRIVE V225W - 32 GB	1	14160	
SMART PHONE	SAMSUNG GALAZY	2	150 000	
NETWORK	Local network installation	1	300 000	
TOTAL 2		7	2,573,688	

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c. Human Resources

Table 7: HUMAN RESOURCES (SOURCE: MERCURIAL 2025)

ROLE	NUMBERS OF DAYS	QUANTITY	COST DAY	PER	TOTAL PRICE (FCFA)
Project Manager	90	01	30 000		2 700 000
Analyst	21	01	25 000		525 000
UI/UX Designer	30	01	10 000		140 000
Programmer	30	01	15 000		450 000
Tester	14	02	10 000		140 000
Margin error	/	/	/		1,475, 000
TOTAL 3					4 840 000

D. GLOBAL ESTIMATION

Table 8: GLOBAL ESTIMATION

TOTAL 1 (FCFA)	TOTAL 2 (FCFA)	TOTAL 3 (FCFA)	OVERALL TOTAL (FCFA)
355,465	2,573,688	4,840,000	7,769,153
SEVEN MILLION SEVEN HUNDRED SIXTY-NINE THOUSAND ONE HUNDRED FIFTY-THREE			

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VI. PROJECT CONSTRAINTS

1. Technical constraint

We possess robust tools that ensure a minimum level of security, extensibility, and excellent scalability for the development of our system. Additionally, adherence to all technical standards during the programming phase will result in improved performance and reduced execution time, making the choice of development technologies crucial.

1. Time Constraint

The project will be completed within a period of 12 weeks (3 months), commencing on the start date of July 08th, 2024

3. Cost constraint

The implementation of our project will necessitate expenses on manpower, materials, and software, amounting to a total of 7,769,153 FCFA.



VII. LIST OF PARTICIPANTS AND DELIVRABLES

1. LIST OF PARTICIPANTS

Table 9: LIST OF PARTICIPANTS

NAME	FUNCTION	ROLE
Mr. TANUE MONET	Follows up interns at the company level	Professional Supervisor
Mr. NDENGE	Follows up student interns at the academic level	Academic Supervisor
FORCHEGODWIN FORBUWING	AICS Student Intern	Student at AICS

2. DELIVRABLES

1. In project management, any component materializing the result of a realization service is called a deliverable. In the case of our project, the deliverables are: A report composed of the following document

- ❖ A User Guide
- ❖ Software Application
- ❖ PowerPoint

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CONCLUSION

In conclusion, this part has come to a close, wherein we have effectively listed and expounded on our goals. The specifications book provided us with the means to present the project's different stakeholders, as well as the prerequisites and tentative schedule essential for its realization. Moving forward, we will directly proceed with the analysis phase, where we will utilize a modelling language and a unified approach to exemplify our system. Furthermore, we will carry out a comparative evaluation of UML and Merise methodologies.

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FILE 3: ANALYSIS PHASE

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PREAMBLE

The primary aim of the analysis phase is to comprehend the user's requirements, define the scope of study and gain a thorough understanding of the system under examination. To accomplish this goal, we will employ the Unified Modelling Language (UML) with the 2 Track Unified Process (2TUP) methodology to analyse the system. Our approach will involve a comparison between UML and MERISE and various other unified processes. Lastly, we will introduce a solution model for our proposed software.

CONTENT

INTRODUCTION

I. COMPARATIVE STUDY OF UML AND MERISE

II. CHOICE OF THE ANALYSIS METHOD

III. JUSTIFICATION OF THE ANALYSIS METHOD

IV. MODELLING OF THE PROPOSED SOLUTION

CONCLUSION

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INTRODUCTION

System development can be thought of as having two major components: System analysis and system design which both help in understanding the details of the existing system or the system to be designed. The analysis and design of information systems has most of the time vocation to allow the creation of databases, which must represent as closely as possible the reality of the field studied thus requiring the use of a design method. This is why our choice will be directed on the UML method as it offers much to developers seeking a user-centered approach and / or a wide scope in design. This part of the report consists of the comparative study of UML and MERISE, unified processes and finally the various diagrams that meet the functional need requirements.

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COMPARATIVE STUDY OF UML AND MERISE

a. MERISE

MERISE stands for “Méthode d’Etude et de Réalisation Informatique pour des Systèmes d’Entreprise”. Although it is prescriptive to some extent, MERISE permits the participation of end users and senior management as well as data processing professionals in its decision cycle. MERISE is a method for designing, developing and carrying out IT projects. The goal of this method is to achieve the design of an information system. The MERISE method is based on the separation of data and processing to be carried out in several conceptual and physical models. The essentials of the approach lie in its three cycles: the decision cycle, the life cycle and the abstraction cycle, which cover data and process elements equally. The separation of data and processing ensures longevity in model. Indeed, the arrangement of data does not have to be often overhauled, while treatments are more frequently.

b. UML

UML (Unified Modelling Language) is a standard notation for the modelling of real world objects as a first step in developing an object-oriented design methodology. Its notation is derived from and unifies the notations of three object-oriented design and analysis methodologies: Grady Booch's methodology for describing a set of objects and their relationships, James Rumbaugh's Object-Modelling Technique (OMT), Ivar Jacobson's approach which includes a use case methodology. Other ideas also contributed to UML, which was the result of a work effort by Booch, Rumbaugh, Jacobson, and others to combine their ideas, working under the sponsorship of Rational Software. UML captures information about the static and dynamic view of a system. UML 2.0 comprises of 13 diagrams which represent the different views of a system. The 13 diagrams can be subdivided into two, Static or structural and Dynamic diagrams. These diagrams include;



1. STATIC OR STRUCTURAL DIAGRAMS

- ❖ Class diagram
- ❖ Object diagram
- ❖ Component diagram
- ❖ Deployment diagram
- ❖ Package diagram
- ❖ Profile diagram

2. BEHAVIORAL PR DYNAMIC DIAGRAMS

- ❖ Use case diagram
- ❖ Activity diagram
- ❖ State machine diagram
- ❖ Sequence diagram
- ❖ Communication diagram
- ❖ Global Interaction diagram
- ❖ Timing diagram

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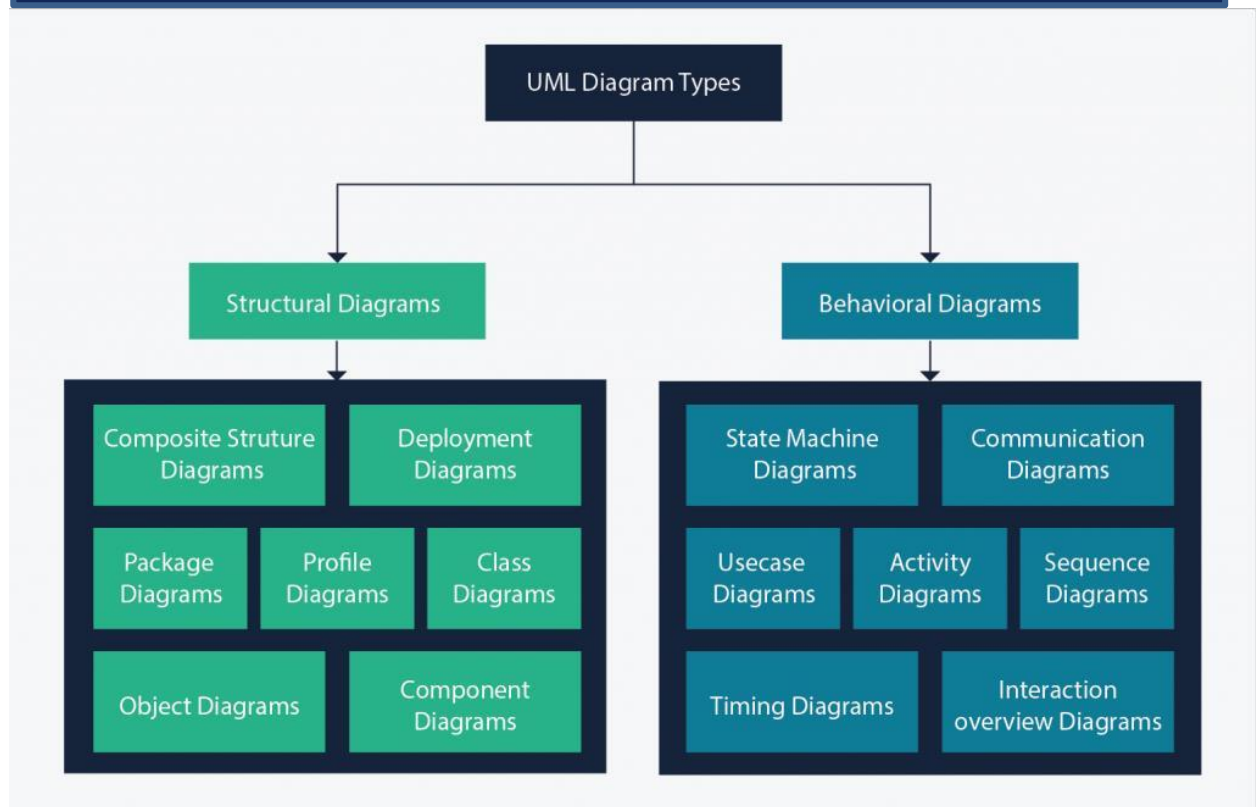


Figure 4: UML 2.5 DIAGRAMS OVERVIEW (SOURCE: [HTTPS://CREATELY.COM/BLOG/DIAGRAMS/UML-DIAGRAM-TYPES-EXAMPLES/](https://createely.com/blog/diagrams/uml-diagram-types-examples/))



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On very important notice is that UML is not a method but a modelling language. As such to give it an approach we need to associate UML to a Unified Process (UP) in order to give our conception a methodology to follow. There exist several Unified Processes but our modelling approach will be the 2TUP (Two-track unified process) which we will use in the course of our project.

Table 10: Comparison between MERISE and UML

MERISE	UML
MERISE is a systemic method of analysis and design of information systems. That is, it uses a system approach.	UML is however not a method but rather an object modelling language to which it is necessary to associate an approach to make it a method. This is the case with the 2TUP method; RUT and XP.
Emphasizes data analysis and modelling using data flow diagrams (DFD), entity-relationship diagrams (ERD), and data dictionaries.	Emphasizes system behaviour and structure using 14 diagram types, such as use case diagram, class diagrams, and activity diagrams.
Suitable for relational databases design and implementation.	Suitable for object-oriented design and implementation.



COMPARATIVE STUDY OF UNIFIED PROCESS

a. Unified Process

A Unified Process is a process of development of software constructed on UML; it is iterative, incremental, centered on architecture, driven by use cases and requirements. Iteration is distinct sequence of activities with a basic plan and evaluation criterion that produces an internal or external output. Either the content of an iteration is improved, or the evolution of the system is evaluated by users. An increment is the difference between two released products at the end of two iterations. Each iteration that the group is capable of integrating the technical environment in order to develop a final product and give users the possibility of having tangible results. Centered on architecture the different models derived during the establishment of system must be reliable and coherent. Driven by use case and requirements enables the clear definition of a users' needs and priorities respectively thereby minimizing the risk of project failure.

b. Two Track Unified Process (2TUP)

2TUP is a unified process which is built on UML and has as objective to bring solution to constraints of functional and technical changes imposed on information systems by strengthening controls on development capacities. It proposes a Y-shaped development life cycle that separates the functional aspect from the technical aspects, and the merging of these two forms the implementation aspect. 2TUP distinguishes therefore two branches: the functional and technical branches, the combination of the result of these two branches forms the third: the realization branch

– where we realize our system. The diagram below illustrates the branches of 2TUP.

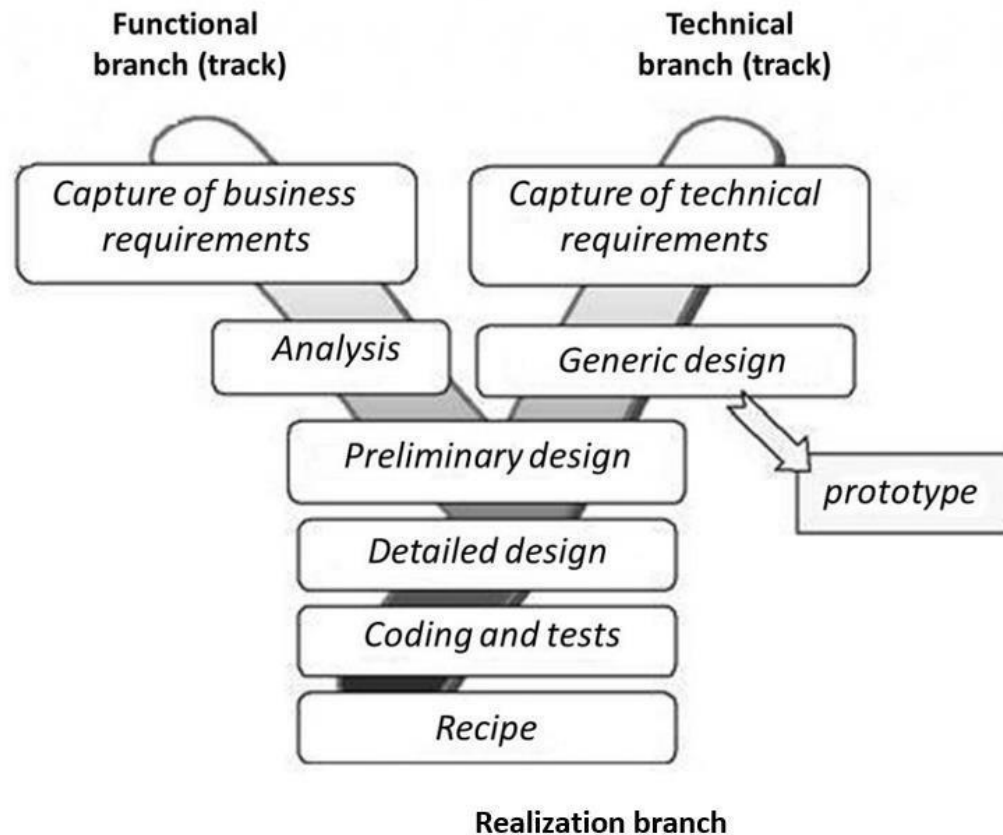


Figure 5: Two TRACK UNIFIED PROCESS (SOURCE:[HTTPS://WWW.MYSCIENCEWORK.COM/OMNISCIENCE/PERVASIVE-MOBILE-HEALTHCARE-SYSTEM-BASEDON-CLOUD-COMPUTING](https://www.mysciencework.com/omniscience/pervasive-mobile-healthcare-system-basedon-cloud-computing))

a) The left branch (functional branch)

It captures the functional needs of a system. This ensures the production of software that meets the needs/requirements of the user. The analysis here consists of studying precisely the functional specification in order to obtain an idea of what the system is going to realize, and its result does not depend on any technology.



b) The right branch (Technical branch)

The technical branch enumerates the technical needs and proposes a generic design validated by a prototype. The technical needs include constraints and choices related to the conception of the system, the tools and equipment as well as the integration constraint with the existing system condition.

c) The middle branch (Realization or Implementation branch)

In this branch, we study the preliminary conception, detailed conception, and documentation of the system. The realization branch supports the following: **Preliminary conception:** This is the most sensitive step of 2TUP as it is the confluence of the functional and technical branch. It is completed when the deployment model, the operating model, the logical model, inter phases and the software configuration model are defined. We have the following diagrams:

- ❖ o Component Diagram
- ❖ o Deployment Diagram
- ❖ o Package Diagram
- ❖ o Composite Structure Diagram



Detailed conception: This is the detailed design of each feature of the system. We have the following diagrams:

- ❖ o Class diagram
- ❖ o Object diagram
- ❖ o Sequence diagram
- ❖ o Timing Diagram

Coding and testing: This is the phase where we program the designed features and test the coded features.

The recipe: Also known as the deliverable is the validation phase of the functions of the developed system



III. JUSTIFICATION OF THE ANALYSIS METHOD

The reason why we chose UML modelling language and the software development process 2TUP instead of many others that exist, include:

- UML is the current standard for programming in an object-oriented language. For this reason, it is widely understood and well known making it easy for a new programmer to join the project and be productive from the very first day.
- UML diagrams allow teams to virtualize how a project is or will be working, and they can be used in any field, not just software engineering. The diagrams will allow teams to virtualize together how a system, or a process will work or did work. It can provide new ideas for how teams have to collaborate to achieve the goal of the workflow process.
- 2TUP is centered around the creation and maintenance of a model, rather than the production of mountain documents.
- 2TUP is user oriented as it permits the development of software that responds to the needs of the users through the study of the user needs.
- 2TUP is iterative and incremental, hence it enables the project team to produce refined amelioration if necessary and easily integrate it in the already existing system.
- 2TUP by permitting the project team identify and test the key functionalities of the system limits the risk related to building the system.



IV. MODELLING OF THE PROPOSED SOLUTION

A. Capture of Functional Needs

The first step of the left (functional) branch of Two Track Unified Process (2TUP) is the capture of the functional needs. At this step, we capture the intended behavior of the system that maybe express as services, tasks or functions the system is required to perform.

1. USE CASE DIAGRAM

a. Definition

Use case diagram shows the functionalities of a system, their interdependencies and how they relate with actors of the system. A use case is a specification of behavior. The main objectives of the use case diagram are:

- Provide a high-level view of the system.
- Identify the functions of the system.

Use case diagrams are completed with a textual description of each use case that is intended to define the use case in greater details.

b. Formalism

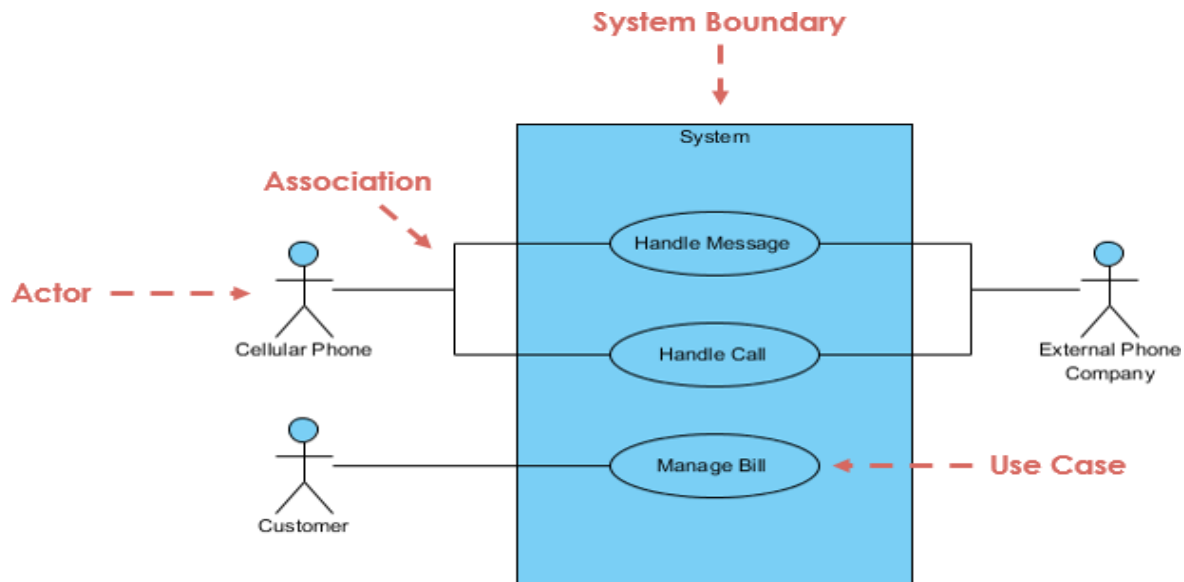
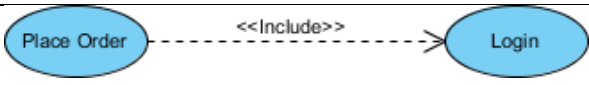
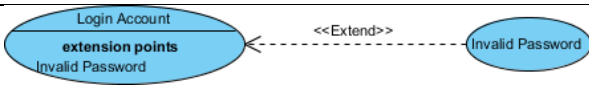


Figure 6: USE CASE DIAGRAM FORMALISM



c. Components of use case diagram

Table 11: COMPONENT OF USE CASE DIAGRAM

Element	Notation	Description
Actors		Represents an entity that directly interacts with the system. The actor is what performs the different possible actions of the system.
Use case		A use case represents a functionality of the system. It is an action that can be performed by an actor.
Association		it indicates that an actor takes part in a use Case.
Include		An inclusion denotes that an included action must be performed before the including action can be performed.
Extend		An extension denotes that an extending action may be performed while an

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		extended action is being performed.
Generalization	 <pre> graph LR A([Store Patient Records (Paper File)]) -- B([Store Patient Records (Computerized File)]) </pre>	This shows that an actor or a use case is a kind of another abstract or concrete actors which can be defined and later specialized using generalization relationship.
System		It is a container of use cases which interact with external actors.



d. Actors of our system

Table 12: ACTORS OF OUR SYSTEM

Actor	Role
Employee	• Login into the platform • Can demand for a leave and justify his absences
Superior or head of department	• Manage employee login into the platform grant leave request and accept justifications
Administrator	- Manage employee and head of department accounts - Approve/reject employee applications and documents - Monitor all activities and resolve disputes - Generate platform performance reports - Assign and manage user roles and permissions and create their account - Create, update, or remove admin accounts - Monitor platform-wide operations for security and compliance - Configure global settings



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e. REALIZE use case diagram

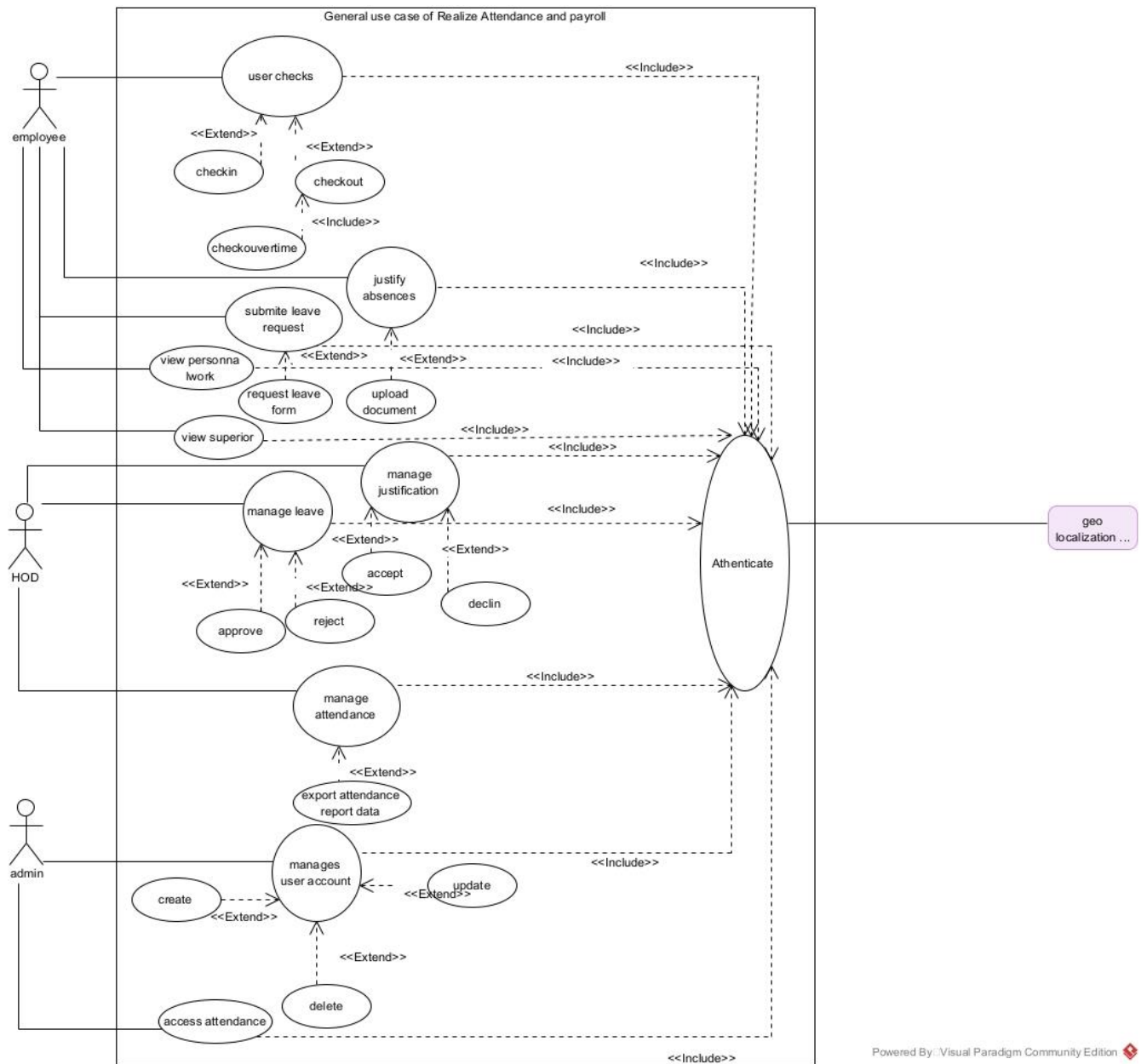


Figure 7: GENERAL USE CASE DIAGRAM



f. Manage account specific use case diagram

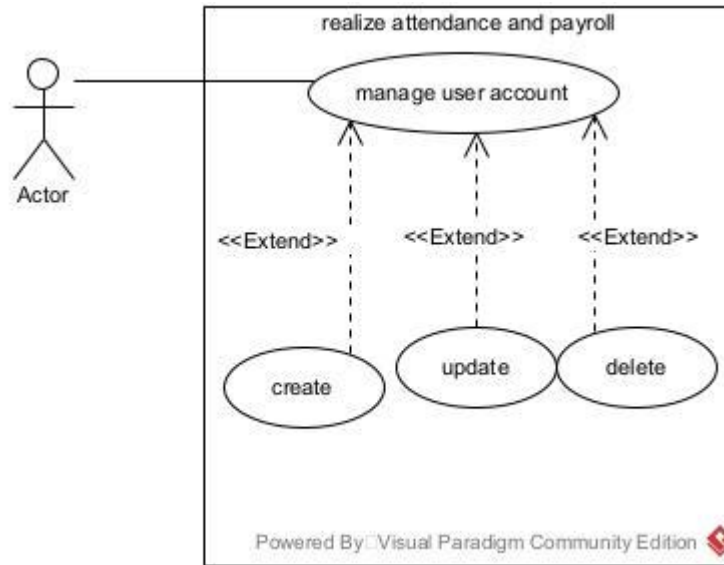


Figure 8:MANAGE ACCOUNT SPECIFIC USE CASE



Textual Description

UML allows the execution of a use case to be described in a textual way, in a form called nominal scenario. A nominal scenario describes in more detail the execution of a use case by an actor until it is successfully completed.

Formalism

A textual description of a use case is represented in the following form

Table 13: TEXTUAL DESCRIPTION FORMALISM

NAME OF THE USE CASE	
Summary	Aim of the use case.
Actors	Users.
Pre-condition	Set of actions that must be completed before the launching of the use case.
Trigger	The event that causes the use case to be initiated.
Nominal scenario	It corresponds to a normal development of a use case.
Alternative scenario	These are variants of the nominal scenario.
Post condition	Set of mechanism that can lead to the end of the use case.



Table 14: TEXTUAL DESCRIPTION OF CREATE ACCOUNT

USE CASE LOGIN ACCOUNT	
Summary	Grant the users access to the application
Actors	Employee
Pre-condition(s)	The app is launched
Trigger	The user clicks on login button
Nominal scenario	1. Click on login account 2. The system displays login form 3. The user fills form and submit 4. The system checks for conformity of the form 5. The system sends a login query 6. The system treats the result returned by the database 7. The system displays successful message
Alternative scenario	8. If step 4 of the nominal scenario the format is not correct, an error message is returned to the user and he returns to step 2 of the nominal scenario. 9. If step 6 of the nominal scenario an error is found, an error message is returned to the user and he returns to step 2 of the nominal scenario
Post-condition of success	The user's account is login
Post condition of failure	An error message is displayed

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2. ACTIVITY DIAGRAM

a. Definition

An activity diagram is a graphical representation of workflows that show the steps needed in the realization of a process; showing the details from a start point to an end point through all decisions and actions that can possibly be performed. Activity diagrams are intended to model both the computational and organizational process. They flow can be sequential, branched or concurrent.

Below is an activity diagram formalism.

b. Formalism

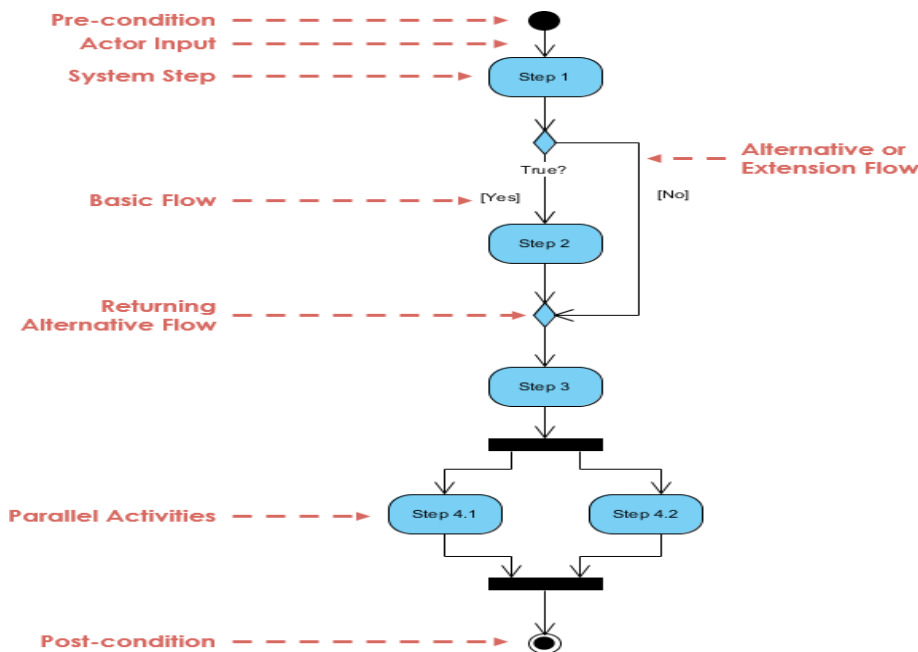
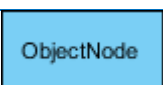
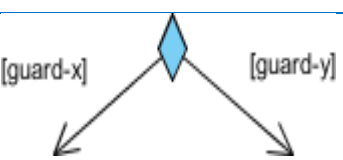


Figure 9: ACTIVITY DIAGRAM FORMALISM

c. Component of an Activity diagram

Element	Notation	Description
Activity		Is used to represent a set of actions
Action		A task to be performed
Activity Edge		An activity edge in a UML activity diagram is a directed Connection between two activity nodes. It represents the flow of control or data between these nodes.
Initial node		Portrays the beginning of a set of actions or activities
Activity Final Node		Stop all control flows and object flows in an activity (or action)
Stop all control flows and object flows in an activity (or action)		Represent an object that is connected to a set of Object Flows
Decision node		Represent a test condition to ensure that the control flow or object flow only

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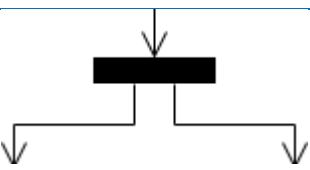
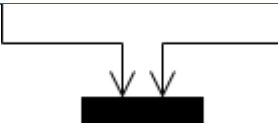
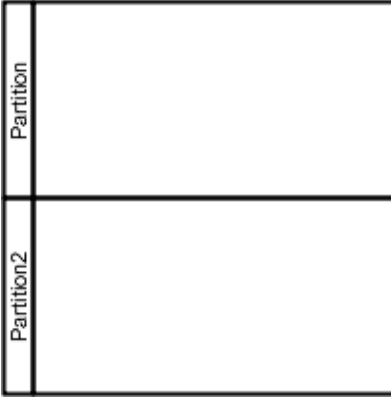
		goes down one path
Merge node		Bring back together different decision paths that were created using a decision- node.
Fork node		Split behavior into a set of parallel or concurrent flows of activities (or actions)
Join node		Bring back together a set of parallel or concurrent flows of activities (or actions).
Swimlane and Partition		A way to group activities performed by the same actor on an activity diagram or to group activities in a single thread

Table 15:COMPONENT OF AN ACTIVITY DIAGRAM

d. Activity diagram of Authenticate

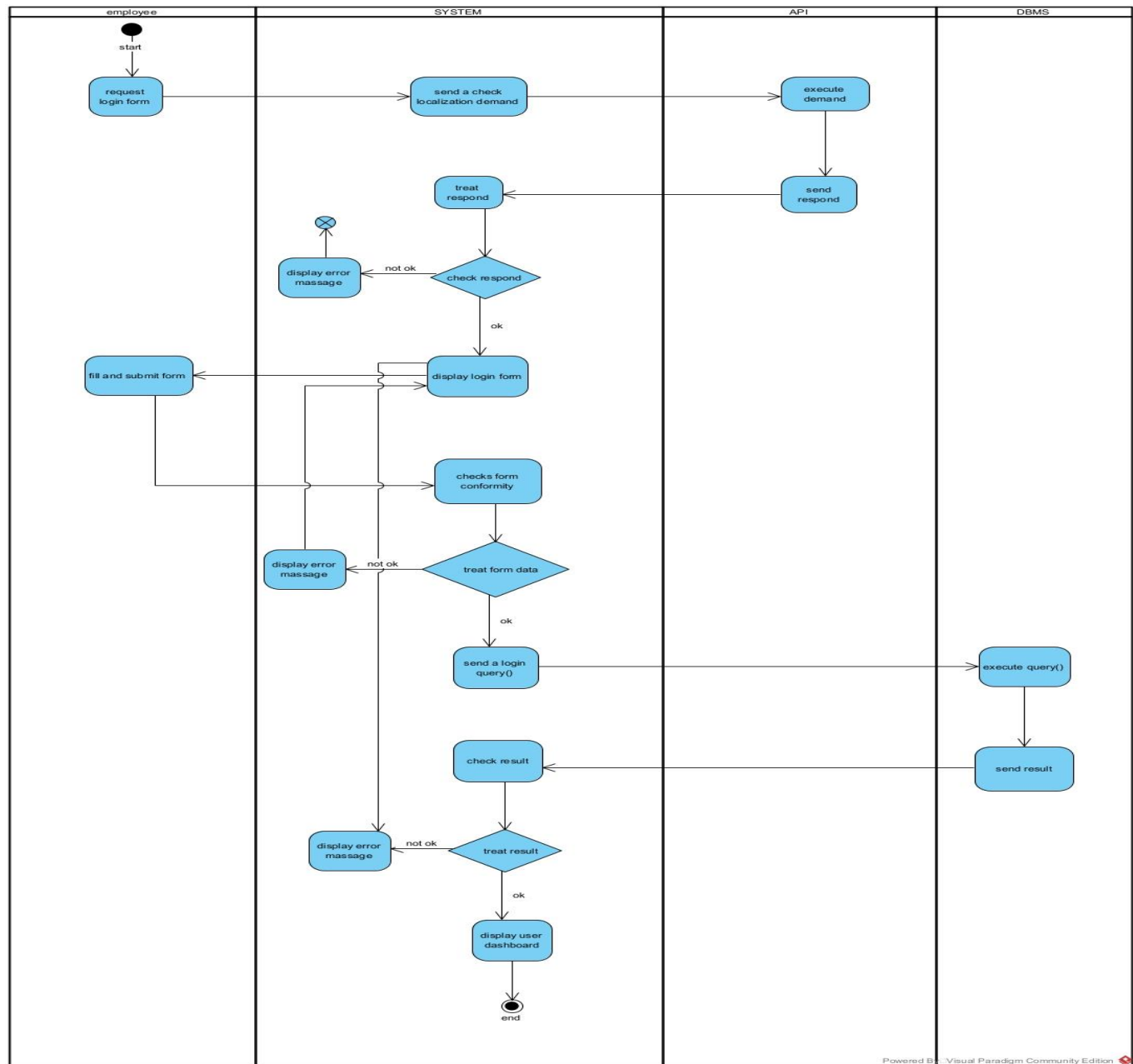


Figure 10: Authenticate activity diagram

e. Activity diagram of leave demand

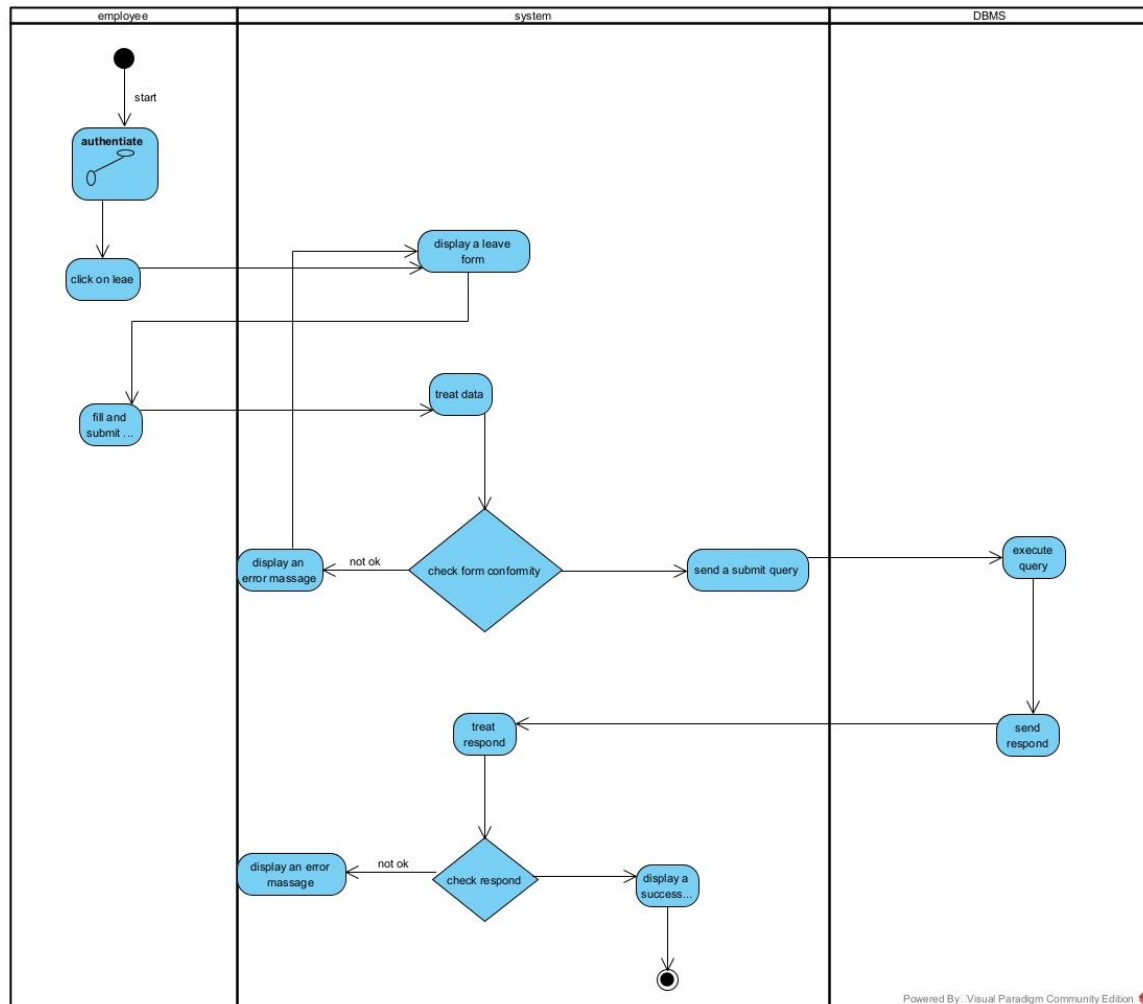


Figure 11: LEAVE DEMAND ACTIVITY DIAGRAM



e. Activity diagram of check-in

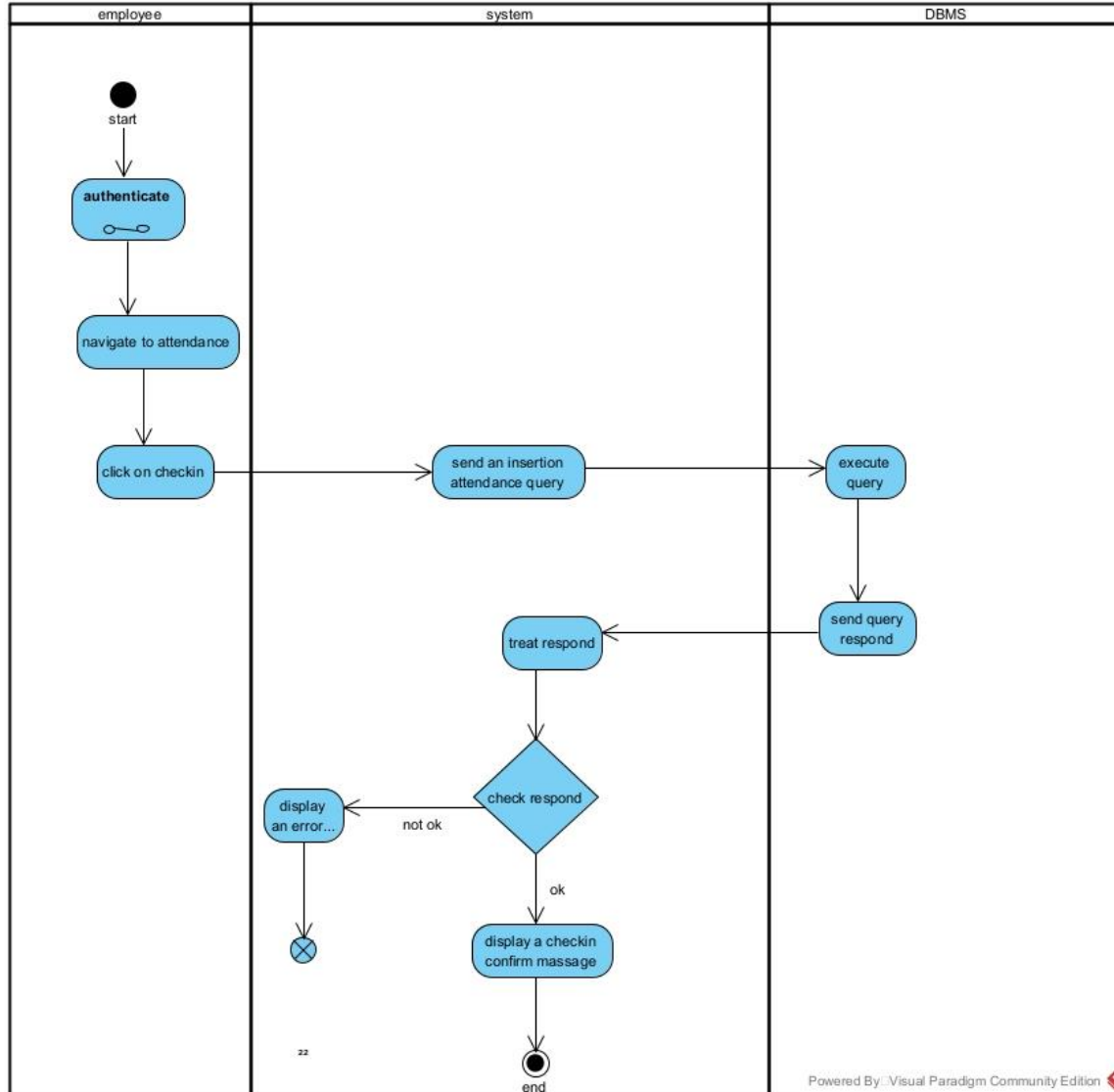


Figure 12: CHECKIN ACTIVITY DIAGRAM



3. COMMUNICATION DIAGRAM

a. Definition

Communication Diagrams model the interactions between objects in a sequence. They describe both the static structure and the dynamic behavior of a system. It is a simplified version of a Collaboration Diagram introduced in UML 2.0. A communication diagram is more focused on showing the collaboration of objects rather than the time sequence.

b. Formalism

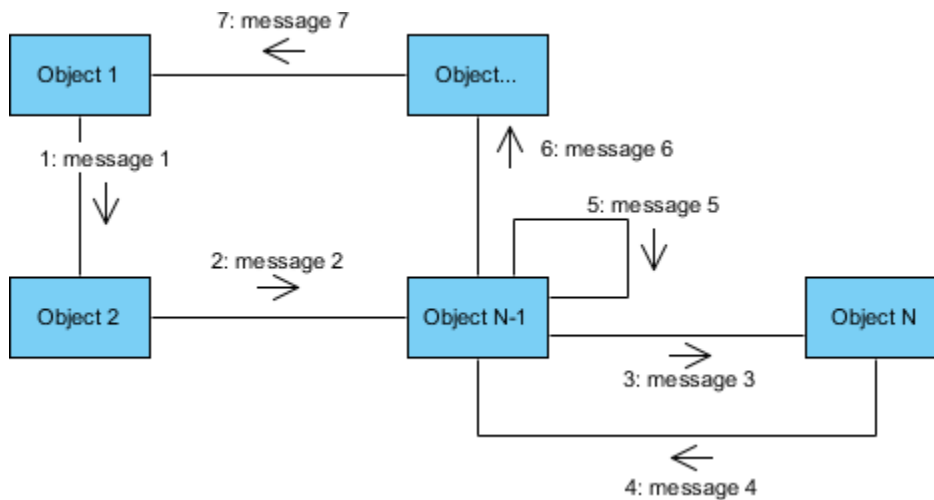


Figure 13: COMMUNICATION DIAGRAM FORMALISM



c. Components of a communication diagram

Table 16: COMPONENT OF A COMMUNICATION DIAGRAM

Element	Notation	Description
Message		Designs a particular communication between lifelines.
Actors		A role play by an entity that interacts with the subjects
Link		It initiates an association it connects two objects together for them to communicate.
Object		An actor represents an individual participant in the interaction conversation



d. Communication diagram of Authenticate

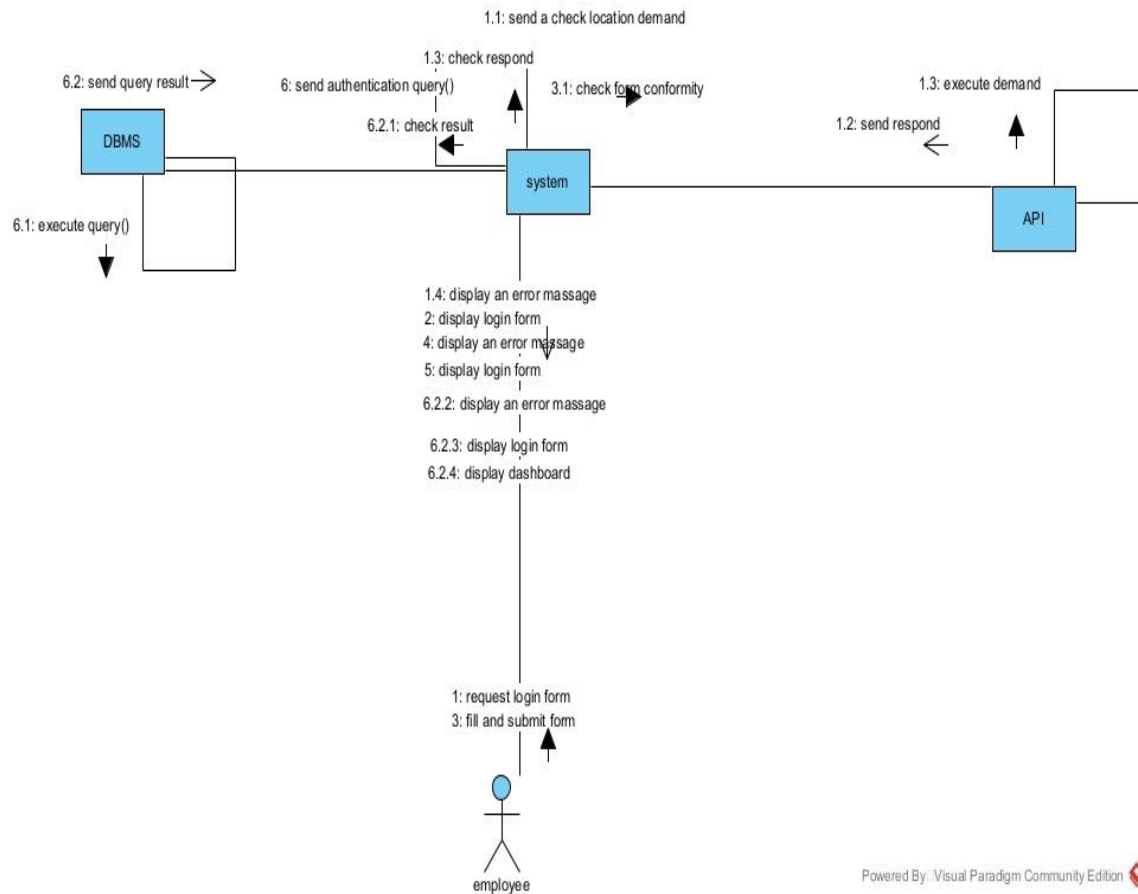


Figure 14: Authenticate Communication diagram

e. Communication diagram of check in

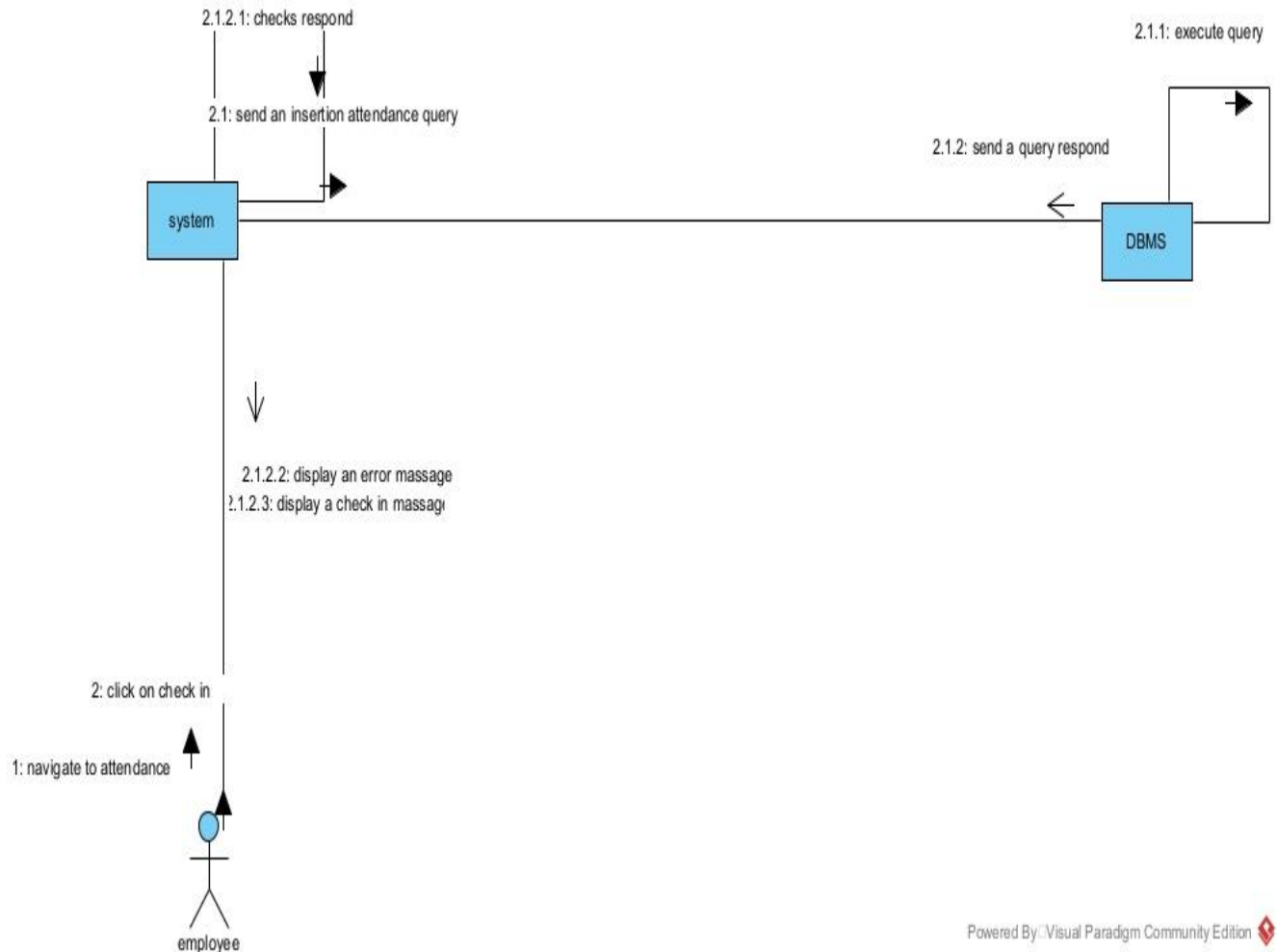


Figure 15: checkin communication diagram

f. Communication diagram of leave demand

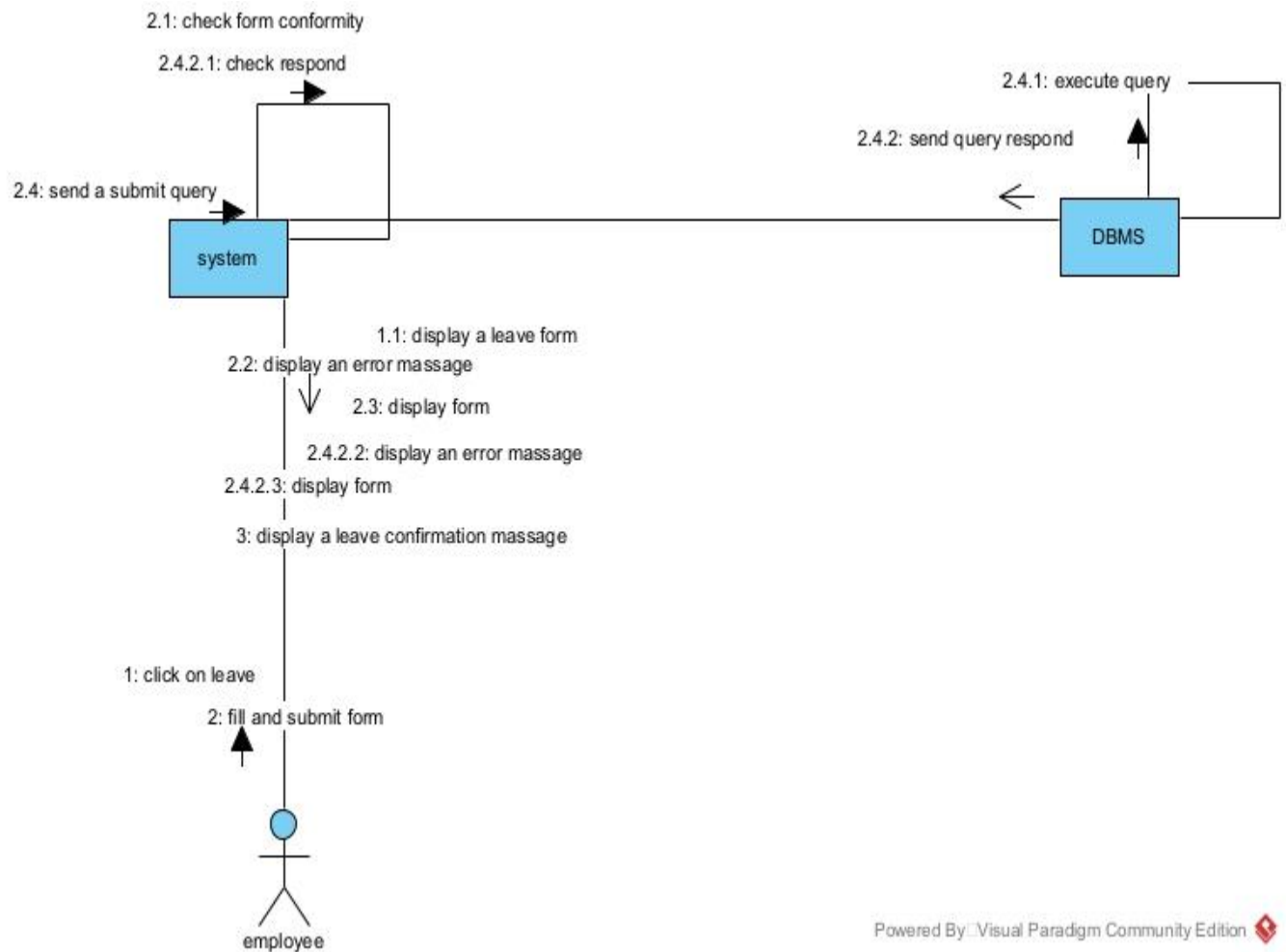


Figure 16: leave demand communication diagram

4. SEQUENCE DIAGRAM

a. Definition

A Sequence diagram describes interactions among classes in terms of an exchange of messages over time. They are also called event diagrams. A Sequence diagram is a good way to visualize and validate various runtime scenarios. These can help to predict how a system will behave and to discover responsibilities a class may need to have in the process of modelling a new system.

b. Formalism

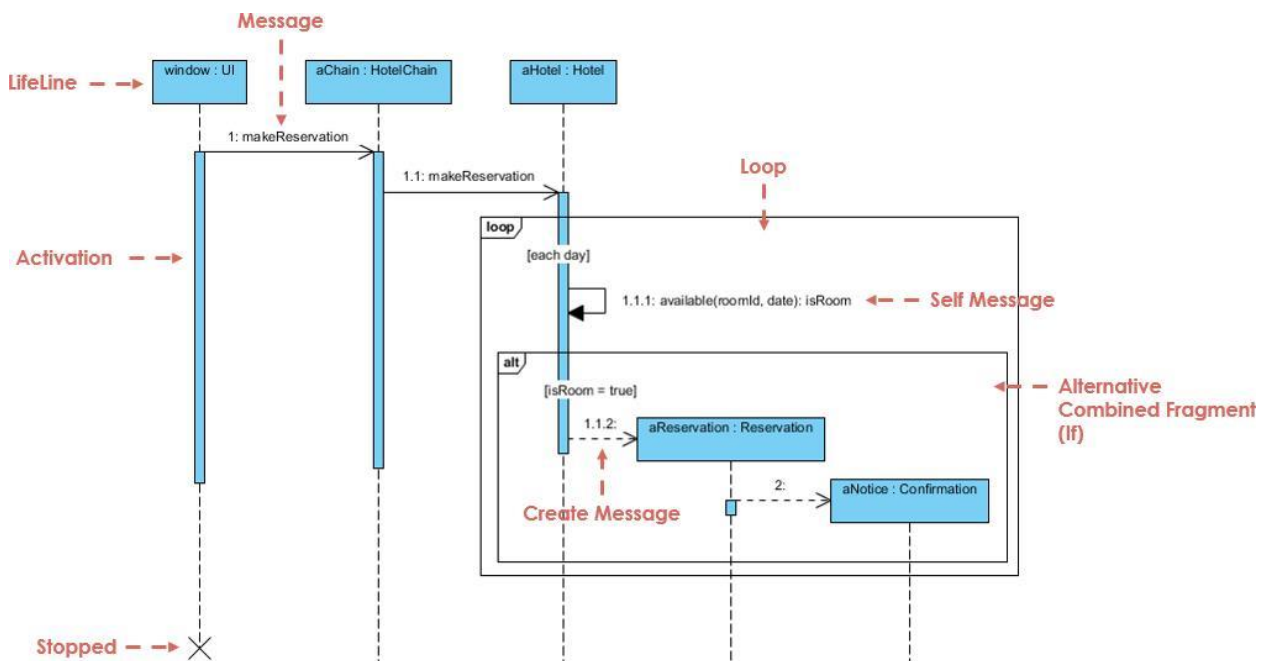
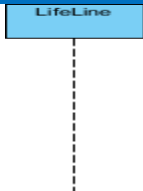
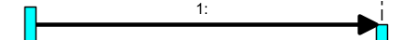
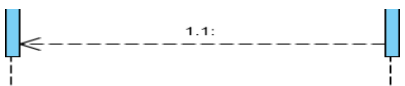


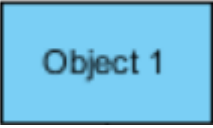
Figure 17: SEQUENCE DIAGRAM FORMALISM

c. Component of sequence diagram

Table 17: COMPONENT OF SEQUENCE DIAGRAM

Element	Notation	Description
Lifelines		They represent rows or objects instances that participate in the sequence being modeled.
Asynchronous Message		It is a message that receives an indirect response.
Synchronous Message		It is a message that sends and want response before it continues a process
Self-Message		Self message is a kind of message that represents the invocation of message of the same lifeline.
Return Message		It represents the response of a message.
Actor		They send and receive message.
Combined Fragment		An articulation interaction of diagram, defined by an operator

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Interaction		and operands.
Object		They send and receive messages

d. Sequence diagram of Authenticate

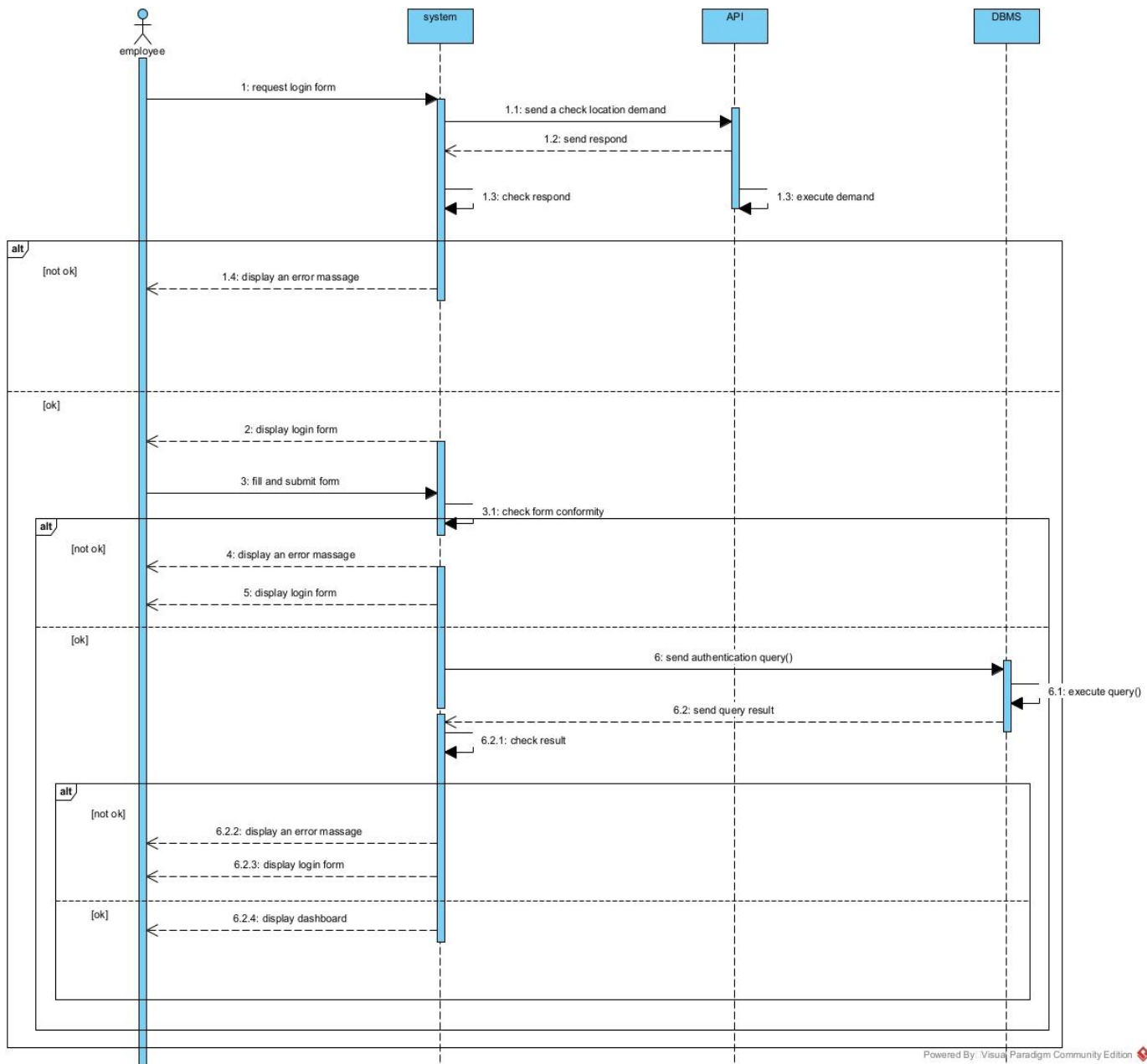


Figure 18: Authenticate Sequence diagram

e. Sequence diagram of check in

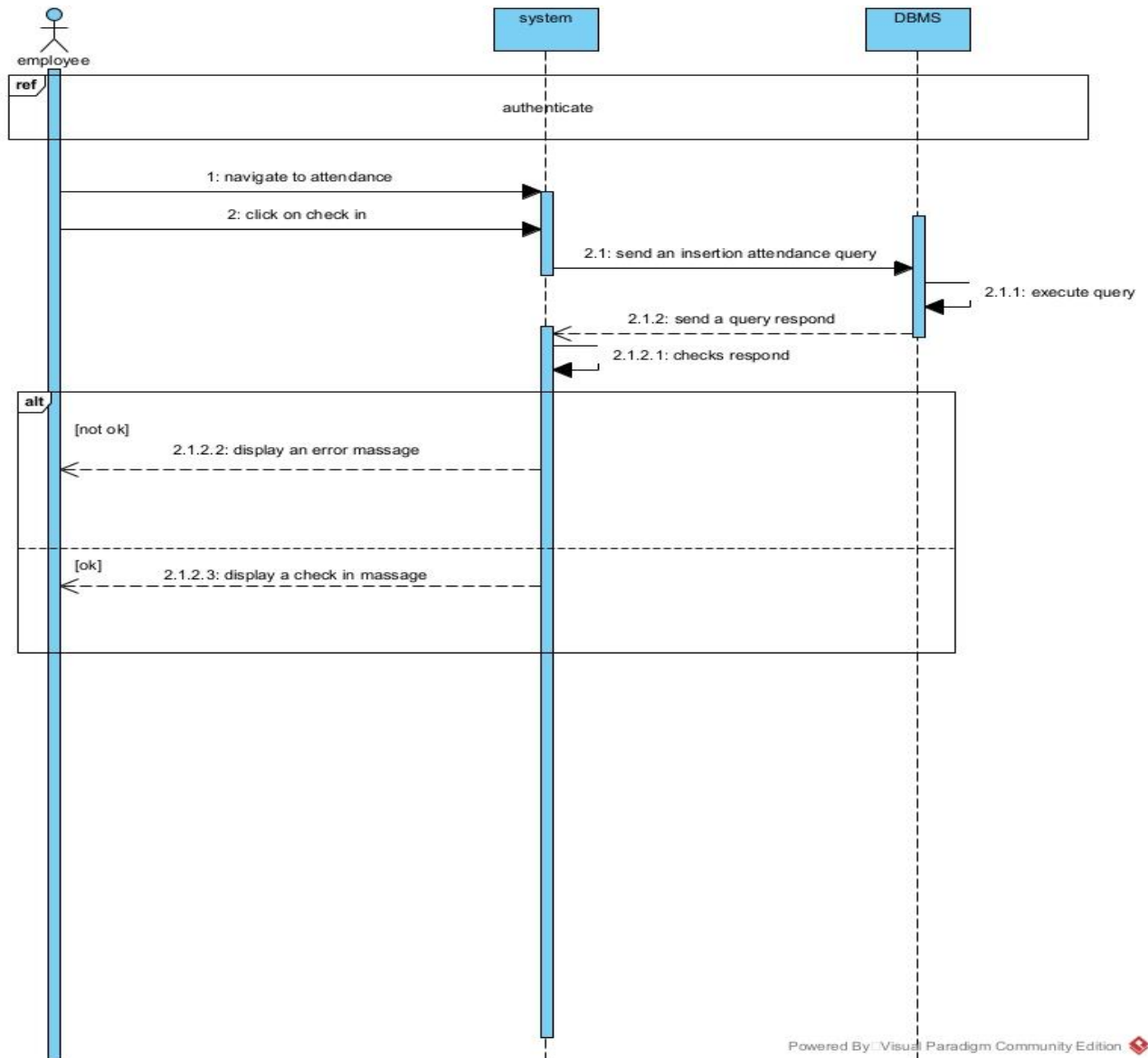


Figure 19: check in sequence diagram



g. Sequence Diagram of leave demand

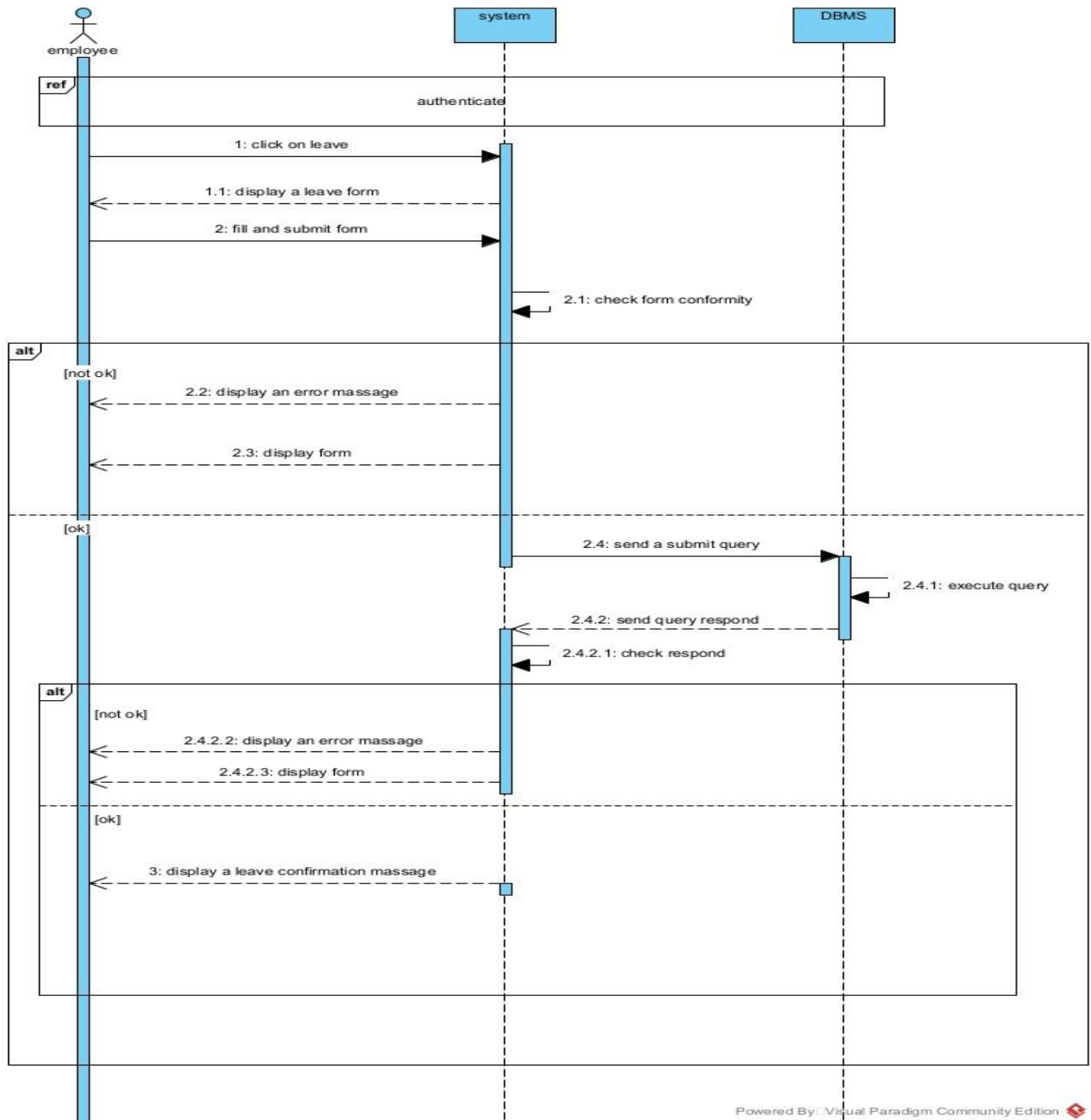


Figure 20: leave demand sequence diagram



CONCLUSION

During the analysis phase, we selected a software development process and modelling language. Next, we elaborated on the functional requirements of our system. We reviewed the use case diagram, which outlines the relationship between actors and the actions they can perform in the system, as well as the communication diagram that illustrates the architecture of the system based on object-oriented programming. We also analyzed the sequence diagram, which delineates the flow of messages between elements in the system. Lastly, we reviewed the activity diagram, which displays the workflow of our system. We will proceed to the conception phase, during which we shall showcase the Technical branch of our system, accompanied by relevant diagrams.

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FILE 4: CONCEPTION PHASE

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PREAMBLE

The conception phase will permit us to present in an orderly manner the components necessary for the good functioning of our software and also the architecture used for the proposed solution. It bridges the gap between the analysis phase and the realization phase.

CONTENT

INTRODUCTION

I. TECHNICAL BRANCH

A. GENERIC BRANCH

B. CAPTURE OF TECHNICAL NEEDS

C. RELATED UML DIAGRAMS

i. CLASS DIAGRAM

ii. STATE MACHINE DIAGRAM

iii. PACKAGE DIAGRAM

CONCLUSION

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INTRODUCTION

The concept stage will outline the precise requirements, characteristics and activities needed to fulfil the operational needs of the suggested system, as defined during the analysis stage. The objective during this phase is to recognize and examine key components (either hardware or software), configuration (network capacity), processes and procedures which are indispensable for the system to achieve its desired outcomes. We will scrutinize several diagrams such as the class diagram, state machine diagram and package diagram.

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TECHNICAL BRANCH

A. GENERIC DESIGN

a. Hardware diagram of the system

The hardware diagram simply shows how the system components of our system are deployed; it shows the positioning of each device into it right proportion.



Figure 21: HARDWARE DIAGRAM

a. High Level Architecture of the System

The high-level architecture diagram provides an overview of the entire system, identifying the main components that would be developed for the product and their interfaces.

B. CAPTURE OF TECHNICAL NEEDS

a. Physical Architecture

The design of the DBMS depends on its architecture. An n-tier architecture partitions on the whole system into related but separated n modules, which can be independently modified, altered, changed or replace. A large amount of data on web servers, personal computers (pc) and others are link with networks with the help of basic client or server architecture. Within the scope of our project, we made use of the 3-tier architecture This architecture separates it tiers from each other based upon the user and the manipulated data in the database. Each layer has a well-defined communication interface, and the evolution of the layer is independent of the other. The 3-tier of our system is made up of:

The client tier, which is also known as our presentation interphase.

Application Tier, which represents our webserver.

The data tier, which represents our DBMS server

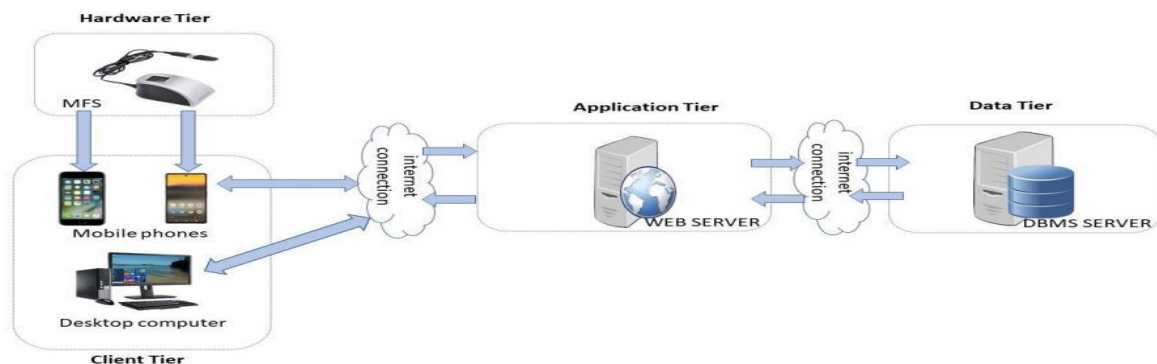


Figure 22: n-tier Architecture (Source: <https://www.pinterest.es/pin/752241943987565648/>)

b. Logical Architecture

Model View controller or MVC as it is popularly called, is a software design pattern for developing application. A model view controller pattern is made up of the following three parts.

- ❖ **Model:** The lowest level of the pattern which is responsible for maintaining data.
- ❖ **View:** This is responsible for displaying all or a portion of data to the user.
- ❖ **Controller:** It handles software codes that controls the interactions between the model and the view.

MVC is popular as it isolates the application logic from the user interface and supports separation of concerns. Here the controller receives all requests for the application then works with the model to prepare data needed by the view. The view then uses the data prepared by the controller to produce a final response. The MVC can be represented as follows:

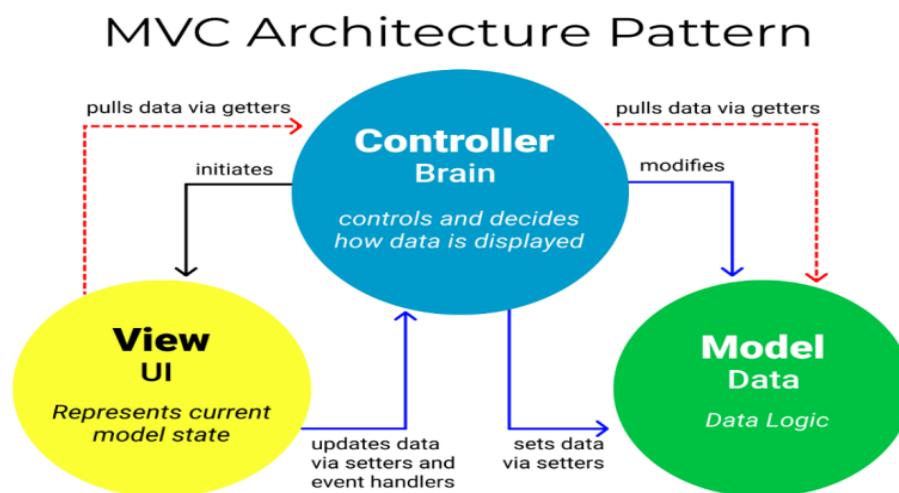


Figure 23: The MVC ARCHITECTURE (SOURCE: [HTTPS://WWW.FREECODECAMP.ORG/NEWS/THE-MODEL-VIEW-CONTROLLER-PATTERN-MVC-ARCHITECTURE-AND-FRAMEWORKS-EXPLAINED/](https://www.freecodecamp.org/news/the-model-view-controller-pattern-mvc-architecture-and-frameworks-explained/))



C. RELATED UML DIAGRAMS

1. CLASS DIAGRAM

a. Definition

A class diagram is a static diagram. It represents the static view of an application. class diagram is not only used for visualizing, describing and documenting different aspect of the system but also for constructing executable code of the software application. Class diagram describes the attribute and operation of a class and constraints imposed on the system. Its purpose is to model the static view of an application.

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b. Formalism

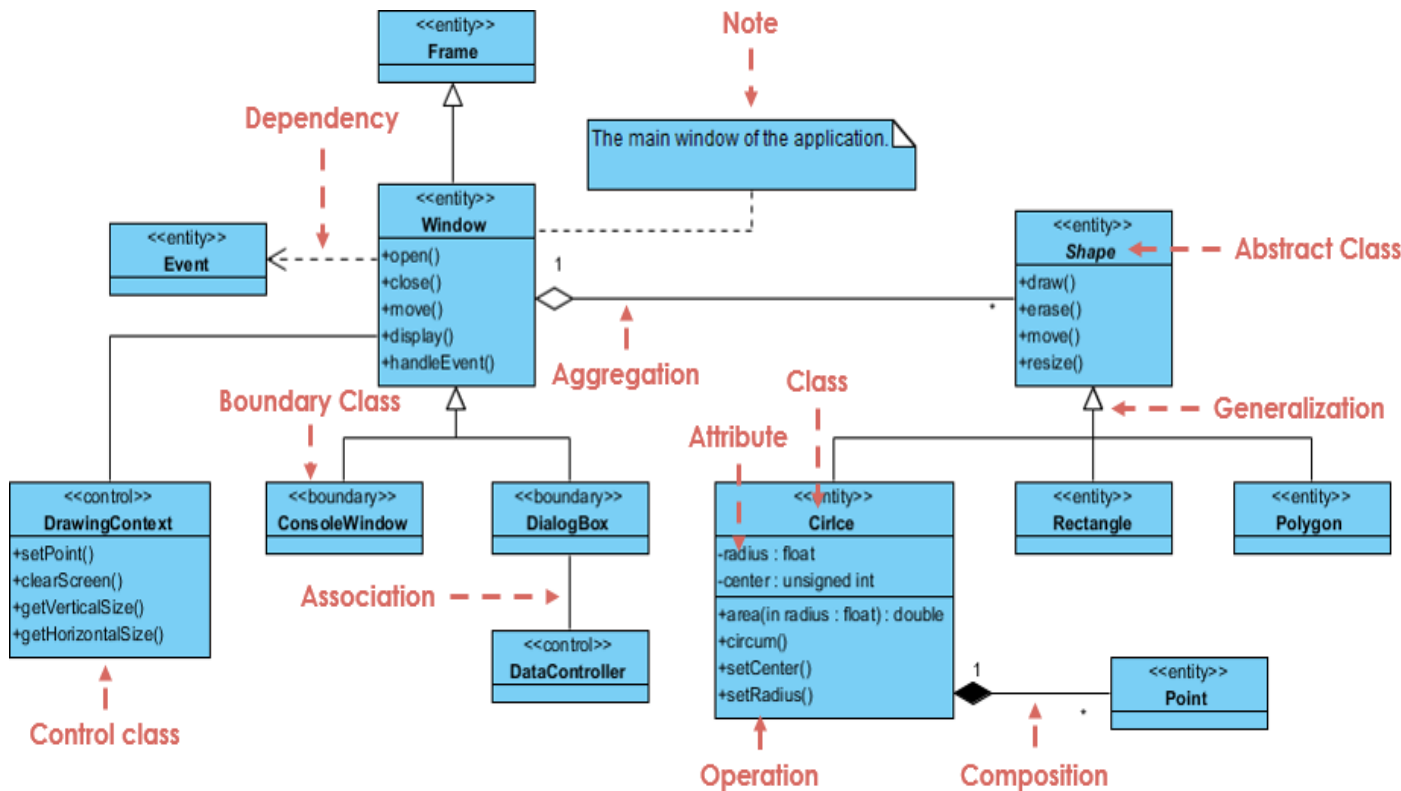
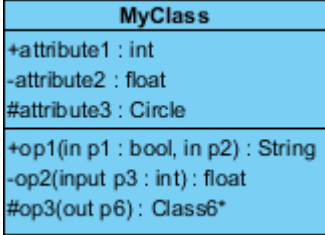
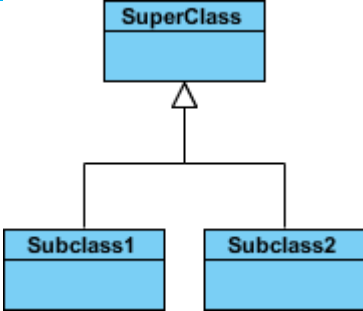
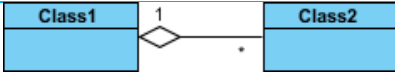
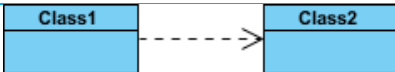
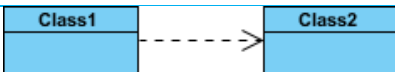


Figure 24: CLASS DIAGRAM FORMALISM

c. Component of a Class diagram

Table 18: COMPONENTS OF CLASS DIAGRAM

Element	Notation	Description
Class	 <pre> classDiagram class MyClass { +attribute1 : int -attribute2 : float #attribute3 : Circle +op1(in p1 : bool, in p2) : String -op2(input p3 : int) : float #op3(out p6) : Class6* } </pre>	A class is an element that defines the attributes and behaviors that an object can generate
Generalization	 <pre> classDiagram SuperClass < -- Subclass1 SuperClass < -- Subclass2 </pre>	it a relationship between a whole thing (called superclass) and a more specific thing (called subclass)
Association	 <pre> classDiagram Class1 --- Class2 </pre>	It is a general type of relationship between elements, it may include cardinality, roles etc.
Aggregation	 <pre> classDiagram Class1 o-- "1" Class2 </pre>	If the parent of the aggregate is deleted, the children are not deleted.
Composition	 <pre> classDiagram Class1 *--> Class2 </pre>	If a parent of a composite is deleted all its parts are deleted with it.
Dependency	 <pre> classDiagram Class1 ..> Class2 </pre>	It existed between two classes, if one changes it may cause the change in the order, but the other way around

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d. realize class diagram

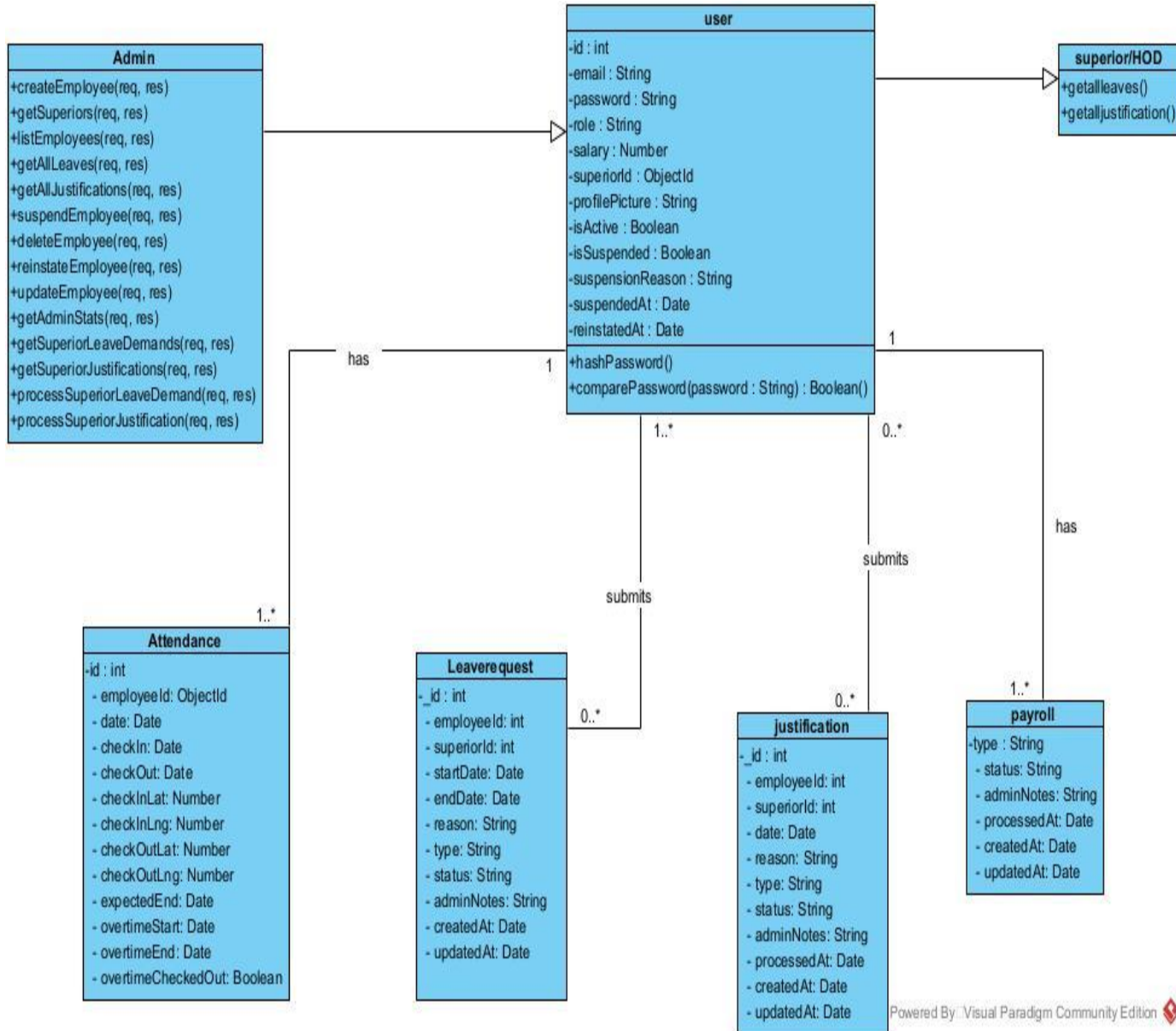


Figure 25: realize class diagram

e. Object diagram

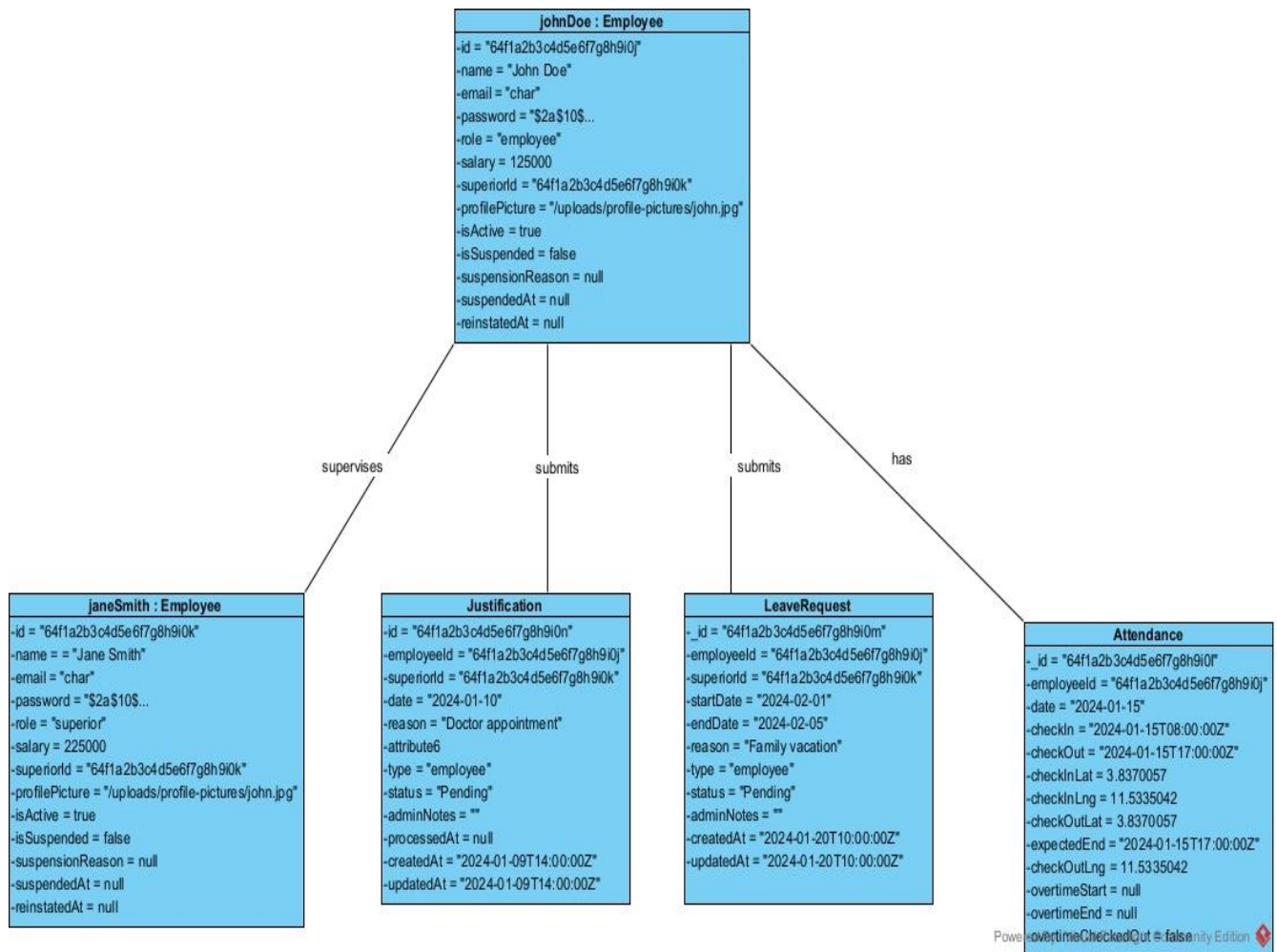


Figure 26:object diagram

2. STATE MACHINE DIAGRAM

a. Definition

A class diagram is a static diagram. It represents the static view of an application. class diagram is not only used for visualizing, describing and documenting different aspect of the system but also for constructing executable code of the software application. Class diagram describes the attribute and operation of a class and constraints imposed on the system. Its purpose is to model the static view of an application.

b. Formalism

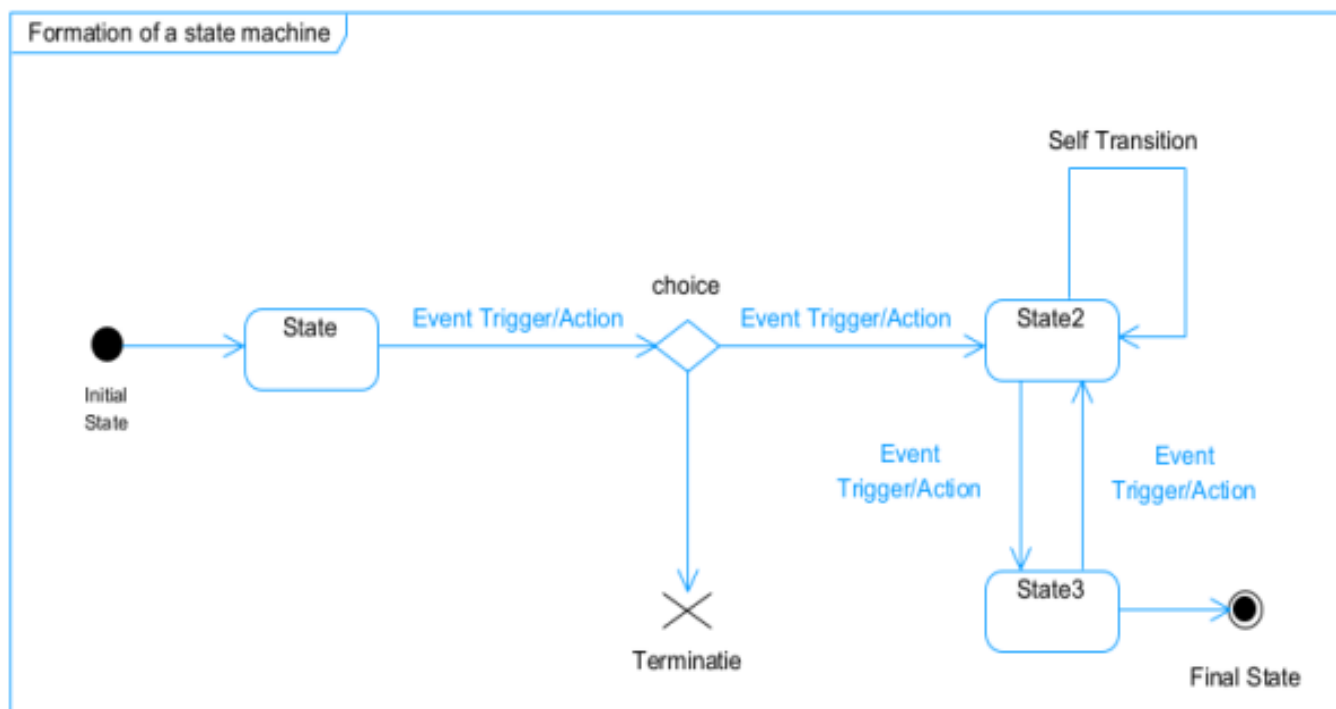
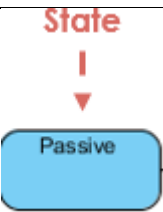
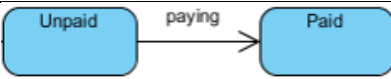
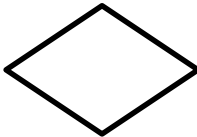


Figure 27: FORMALISM OF A STATE MACHINE DIAGRAM

c. Component of a state machine diagram

Table 19: COMPONENT OF STATE MACHINE DIAGRAM

Element	Notation	Description
State		Models a situation during which a certain invariant condition holds.
First (Initial State)		It represents a default vertex, that is, a source for a single transaction to the default or composite state.
Final state		A state specifying that the Enclosing region is complete
Transition		A direction relation between a source and a target vertex
Choice pseudo state		A diamond symbol that indicates a dynamic condition with branched potential results
Terminate		Implies that the execution of a state by means of its context is terminated.
Diagram overview		A placeholder for the linked states in a state machine diagram.

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DESIGN AND IMPLEMENTATION OF A WEB APPLICATION FOR EMPLOYEE ATTENDANCE TRACKING AND PAYROLL MANAGEMENT (case study realize enterprise)



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d. Create account state machine diagram

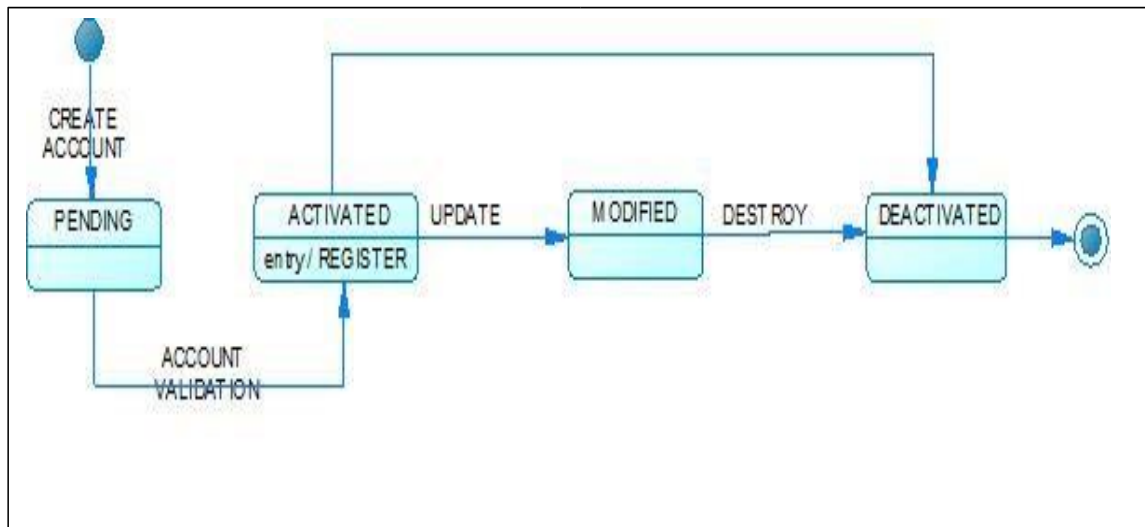


Figure 28:create account state machine diagram

3. PACKAGE DIAGRAM

a. Definition

This is a structural diagram used to show the organization and arrangement of various model elements in the form of packages. A package diagram is the grouping of related uml elements such as classes, diagrams or eve other packages.

b. Formalism

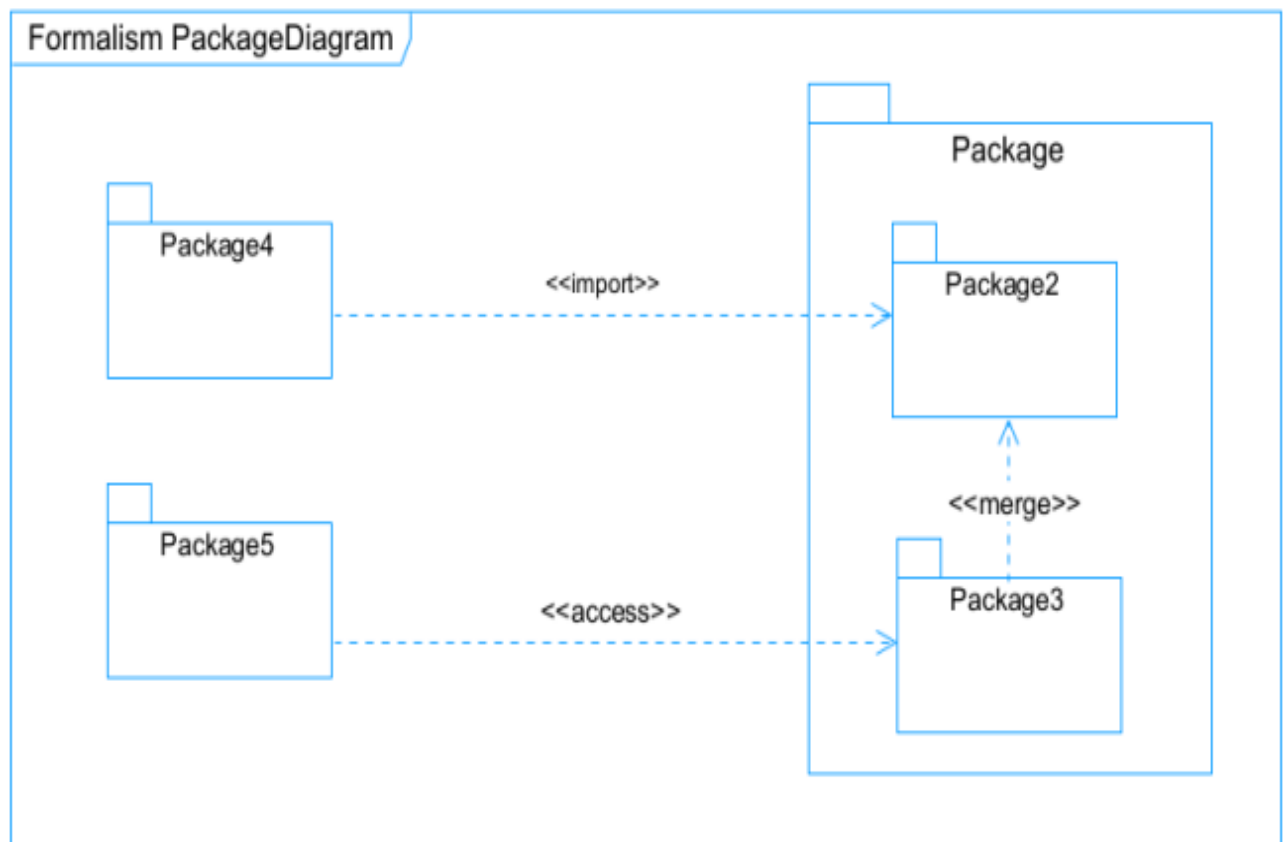


Figure 29: PACKAGE DIAGRAM FORMALISM

c. Component of package diagram

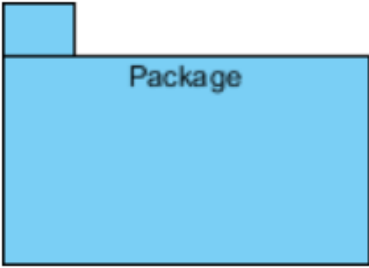
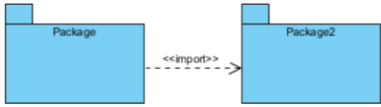
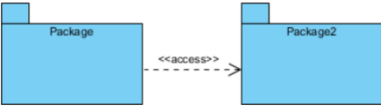
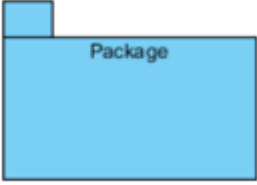
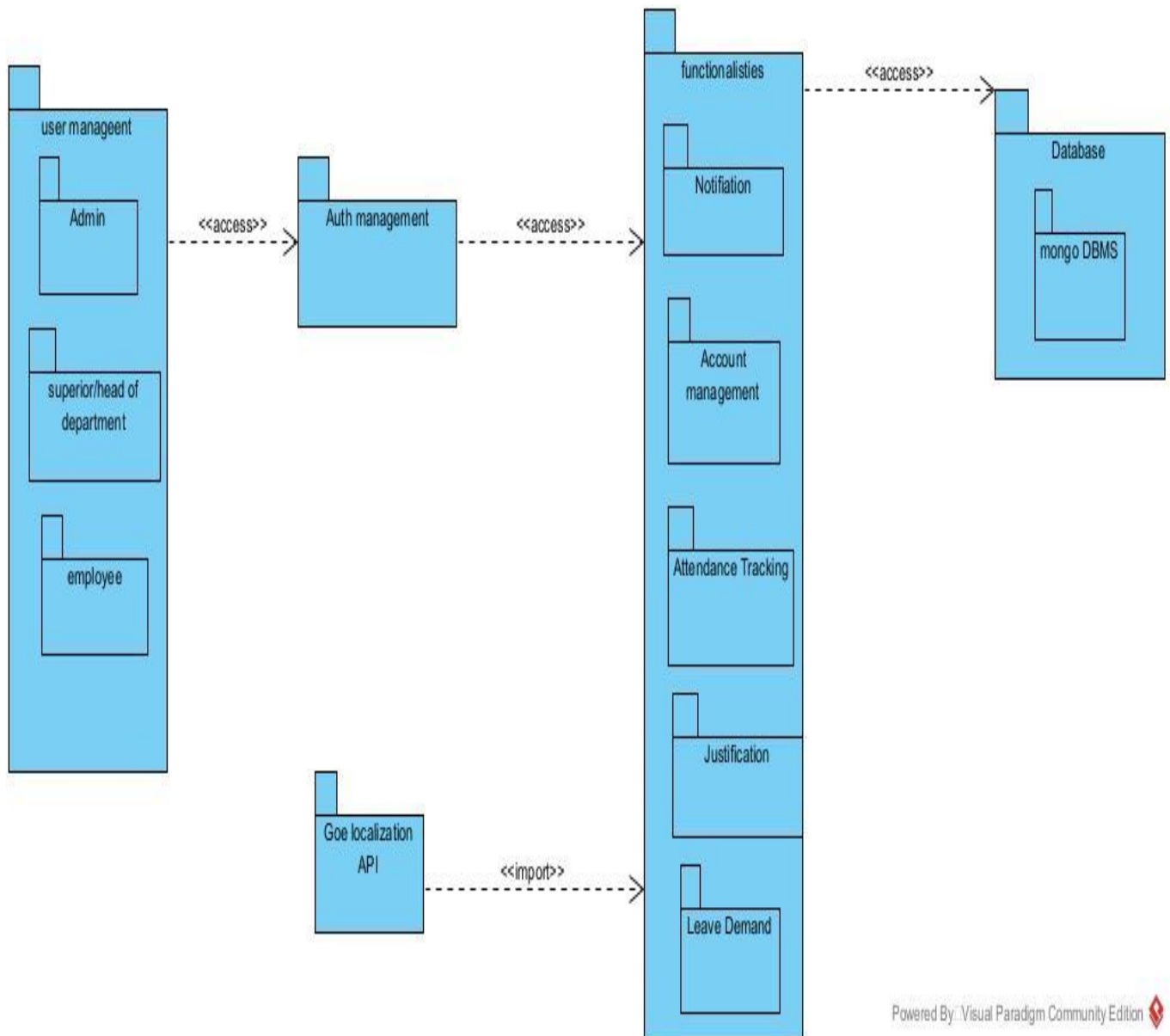
Package	Notation	Description
Package		A package is a name space use to group related elements; it is a mechanism used to group elements into a better structure in a system.
Package import		A relationship Indicate that, functionality has been imported from one package to another.
Package access		A relationship Indicates that one package requires assistance from the function of another package.
Package merge		It is a relationship which shows that, the functionality of two packages are combines to a single function.

Figure 30: Component of package diagram

d. Package diagram



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Figure 31:package diagram

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CONCLUSION

We presented the structure of our system and its integration into the existing technical architectures. In the initial stage, we aimed to investigate various aspects of our system. Primary actions included assessing the technical limitations, identifying the system's components with their classification and deployment onto relevant deployment targets. We then concluded this phase by evaluating the interactions between different components and participants of our system. Moving forward, our report will now progress to the implementation phase, which will entail concentrating on executing our system-related elements.

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FILE 5: REALIZATION PHASE

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In this phase, we aim to implement our solution straightforwardly by relying on the analysis and conception phases. Additionally, we will showcase the component and deployment diagrams.

CONTENT

INTRODUCTION

I. DEPLOYMENT DIAGRAM

II. COMPONENT DIAGRAM

CONCLUSION

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1. DEPLOYMENT DIAGRAM

a. Definition

Deployment diagram is a structural diagram used to visualize the topology of the physical components of a system, where the software is deployed. They consist of nodes and their relationship. It is related to the component diagram because the components are deployed using the deployment diagram. A deployment diagram consists of nodes. Nodes are nothing but physical hardware used to deploy the application.

b. Formalism

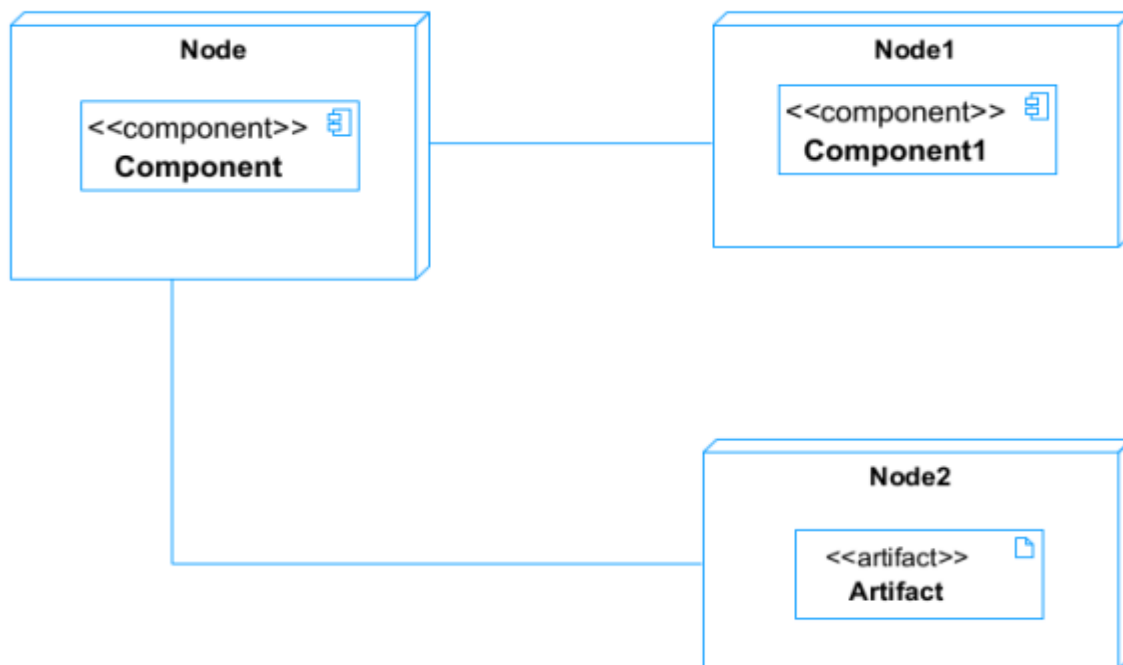
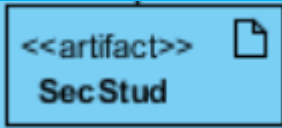
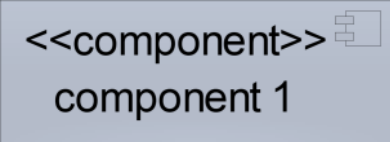


Figure 32: FORMALISM OF DEPLOYMENT DIAGRAM



Element	Notation	Description
Node		A hardware or software object, shown by a three-dimensional box
Artifact		An artifact is a major product, which is produced or used during the development of a software. E.g diagrams, data models, setup scripts
Component		It represents a modular part of a system that encapsulates its content and whose manifestation replaceable is within its environment.
Association		An association helps to connect two nodes together which permits them to communicate together

d. Deployment diagram

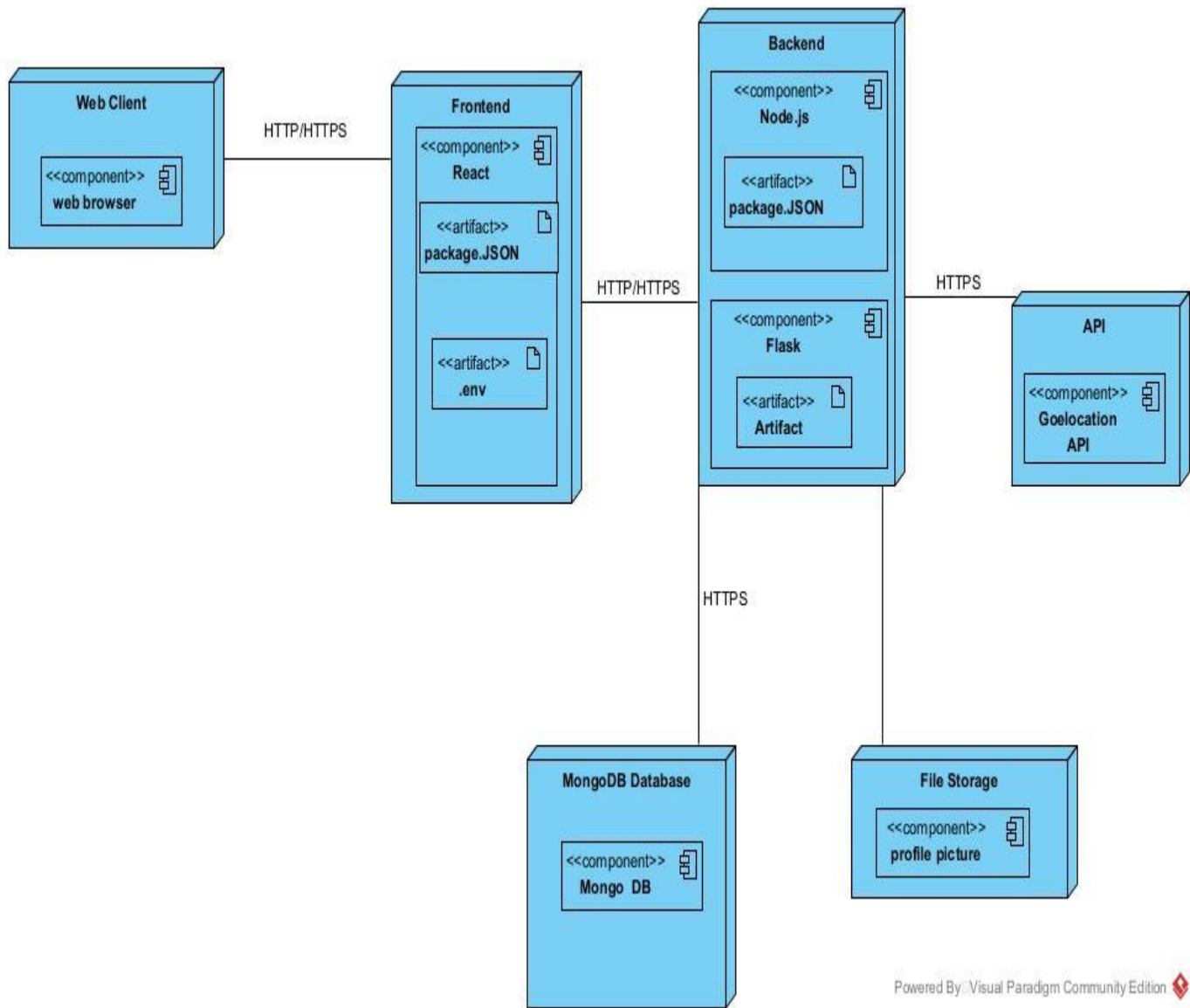


Figure 33: deployment diagram

2. COMPONENT DIAGRAM

a. Definition

Component diagrams are used to model the physical aspect of a system. Now the question is what are this physical aspect? They are elements such as Executable, libraries, files, document etc. which resides in a node. The component diagram does not describe the functionality of the system, but it describes the components used to make those functionalities.

b. Formalism

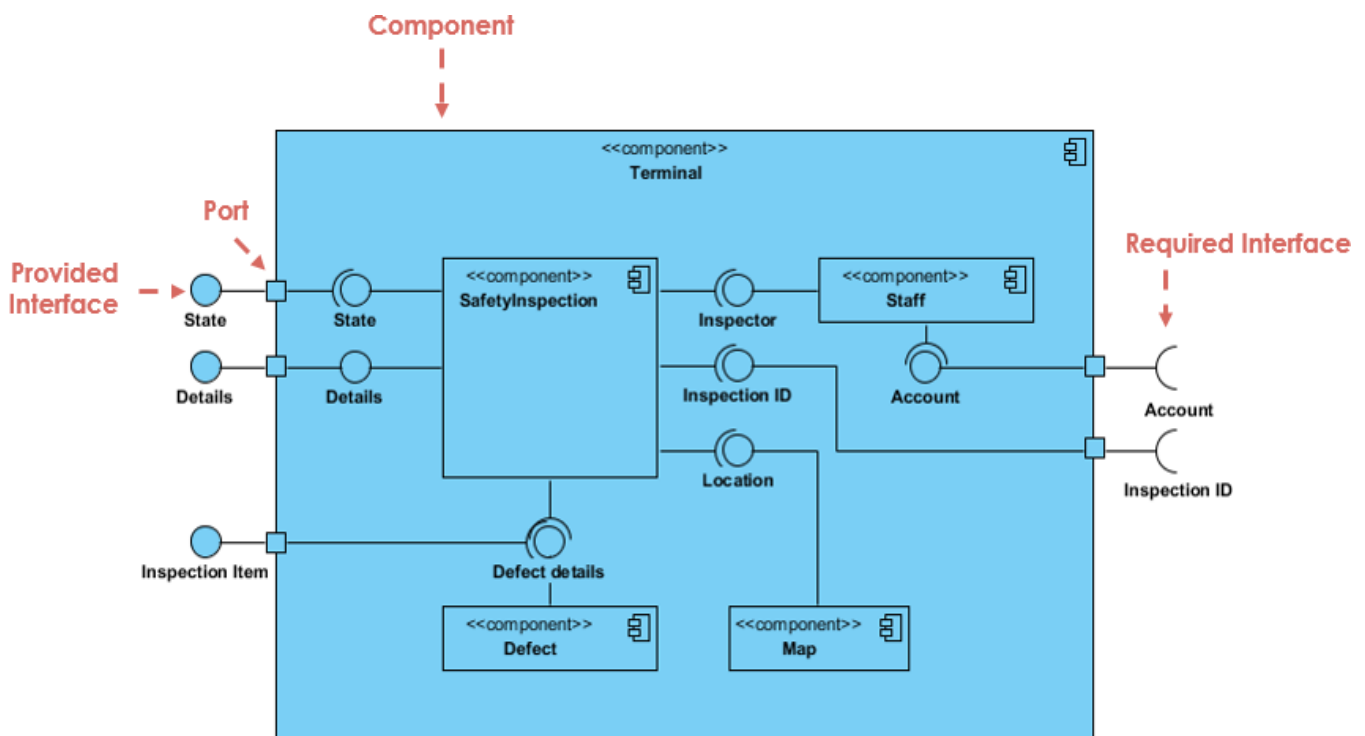
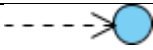


Figure 34: COMPONENT DIAGRAM FORMALISM



c. Component of Component diagram

Table 20: Component of deployment diagram

Element	Notation	Description
Dependency		A dependency is a relationship that signifies that a single or a set of model elements requires other model elements for their specification or implementation.
Port		A port is often used to help expose required and provided interfaces of a component.
Interface		It is a point of interaction between a component and its environment or other components. It specifies the services that a component provides or requires from other components
Component		It represents a self-contained and replaceable unit that interacts other with components via provided and required interfaces

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d. Component diagram

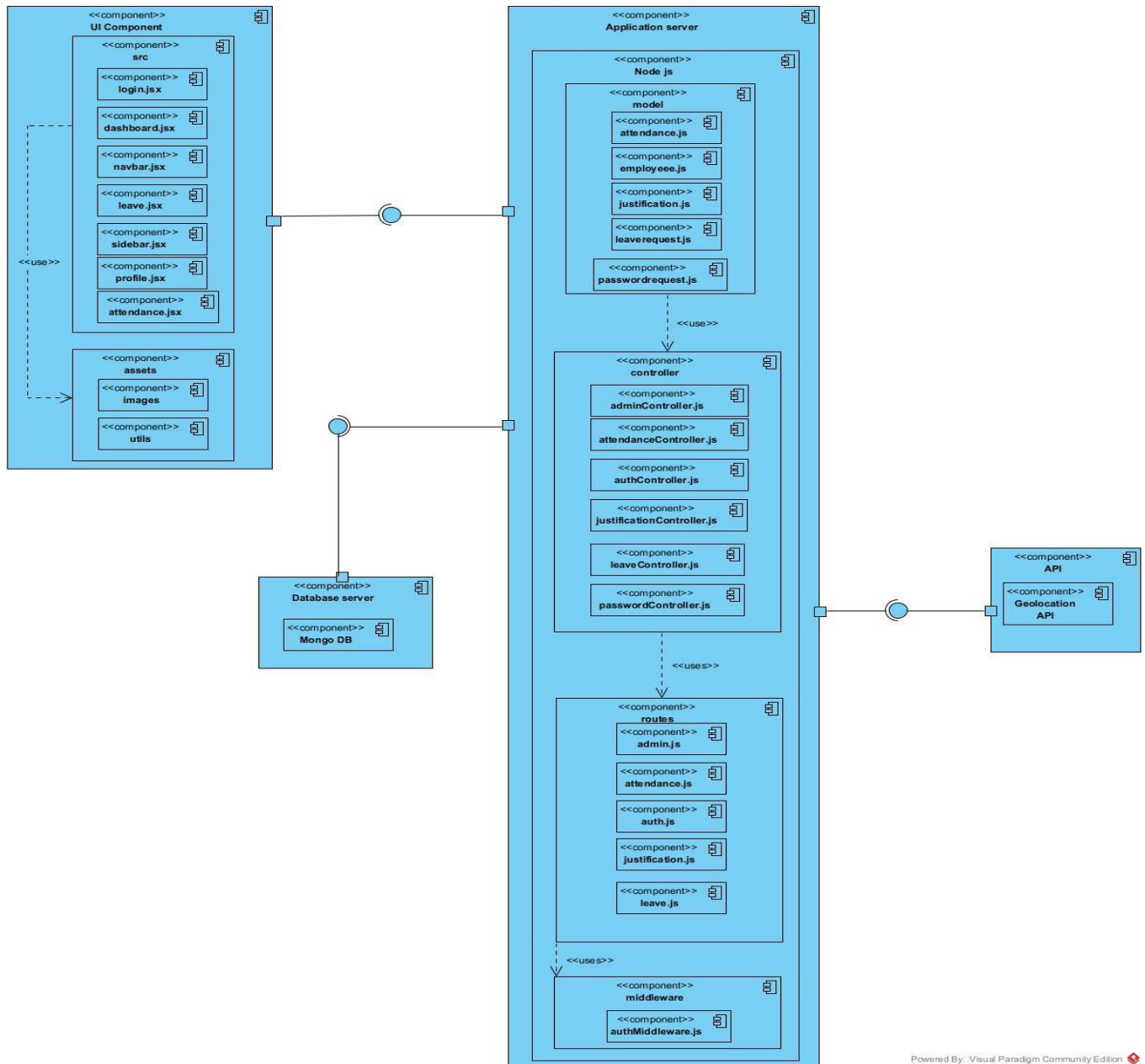


Figure 35: component diagram



CONCLUSION

During our realization phase, we successfully developed and implemented our application. To achieve this, we utilized our analysis and conception phase. We also created deployment and component diagrams which illustrate the modules, files, and assets, as well as how different elements interact with each other within our system. Our next step is to move onto the functionalities testing phase, where we will assess the efficacy of the various modules present in our app for different users.

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FILE 6: TEST OF FUNCTIONALITIES

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PREAMABLE

In this report, we will explore the different aspects of the test of functionalities and its significance in the software development life cycle.

CONTENT

INTRODUCTION

I. TEST OF FUNCTIONALITIES

II. TEST SHOWCASES

CONCLUSION

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INTRODUCTION

The test of functionalities plays a crucial role in ensuring the quality and reliability of software systems. By thoroughly examining each functionality and validating its behavior, developers can identify potential issues early in the development process. This helps in reducing the overall cost and effort required for bug fixing and maintenance.

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TEST OF FUNCTIONALITIES

1. What is Unit Testing

Unit testing involves testing the smallest parts of an application, such as functions or methods, independently from the rest of the application. The primary goal is to validate that each unit of the software performs as expected.

2. Unit Testing Importance

Unit testing ensures that each component of the application works independently and correctly. It helps catch bugs early in the development process and guarantees that future changes don't break existing functionality.

3. Testing Strategy

We implemented unit testing to isolate and test individual parts of the app, specifically the API endpoints and backend services. Unit testing provides fast feedback on code quality and behavior.

4. Tools Used

- Jest: Testing framework used for running unit tests.
- Supertest: For testing API requests.
- Mockingoose: Used in some instances to mock MongoDB models for tests.

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5. Key Functionalities Tested

a. User Leave demand

- Tests ensure that users can successfully demand leave with valid details.
- Handles test cases where invalid inputs or missing fields cause demand failure.

b. User Authentication

- Testing confirm that users can login with correct credential.
- Includes scenarios where login fails due to invalid credentials (e.g., wrong password).

c. User Creation

- Tests verify that user is created successfully with all required information.
- Includes test cases where an error is returned if a user creation with the same credentials.

TEST SHOW CASES

a. Passing Unit Tests

- User login: Demonstrates successful user login with valid data.

```
Auth Controller - Login Tests
Successful Admin Login
password (plain text from req): validpassword123
  1) should successfully login admin user
password (plain text from req): validpassword123
  2) should return password warning for admin with temporary password
Successful Employee Login
password (plain text from req): validpassword123
  3) should successfully login employee with valid location
password (plain text from req): validpassword123
  4) should successfully login employee with location from IP
Successful Superior Login
```

Figure 36: User login test

- User Authentication: Shows successful login and token generation when valid credentials are provided

```
backend > test > JS auth.test.js > ...
const chai = require('chai');
const sinon = require('sinon');
const { expect } = chai;
const { mockEmployee } = require('../setup');

// Import the auth controller (we'll need to modify it slightly for testing)
const authController = require('../controllers/authController');

describe('Auth Controller - Login Tests', () => {
  let req, res, mockUser;

  beforeEach(() => {
    // Reset all stubs
    sinon.reset();

    // Mock request and response objects
    req = {
      body: {},
      ip: '127.0.0.1'
    };
    res = {
      status: sinon.stub().returnsThis(),
      json: sinon.stub()
    };

    // Mock user data for different roles
    mockUser = {
      _id: '507f1f77bcbf86cd799439011',
      name: 'Test User',
      email: 'test@example.com',
      password: 'hashedpassword123',
      role: 'employee'
    };
  });
});
```

Figure 37: Authentication test code snippet

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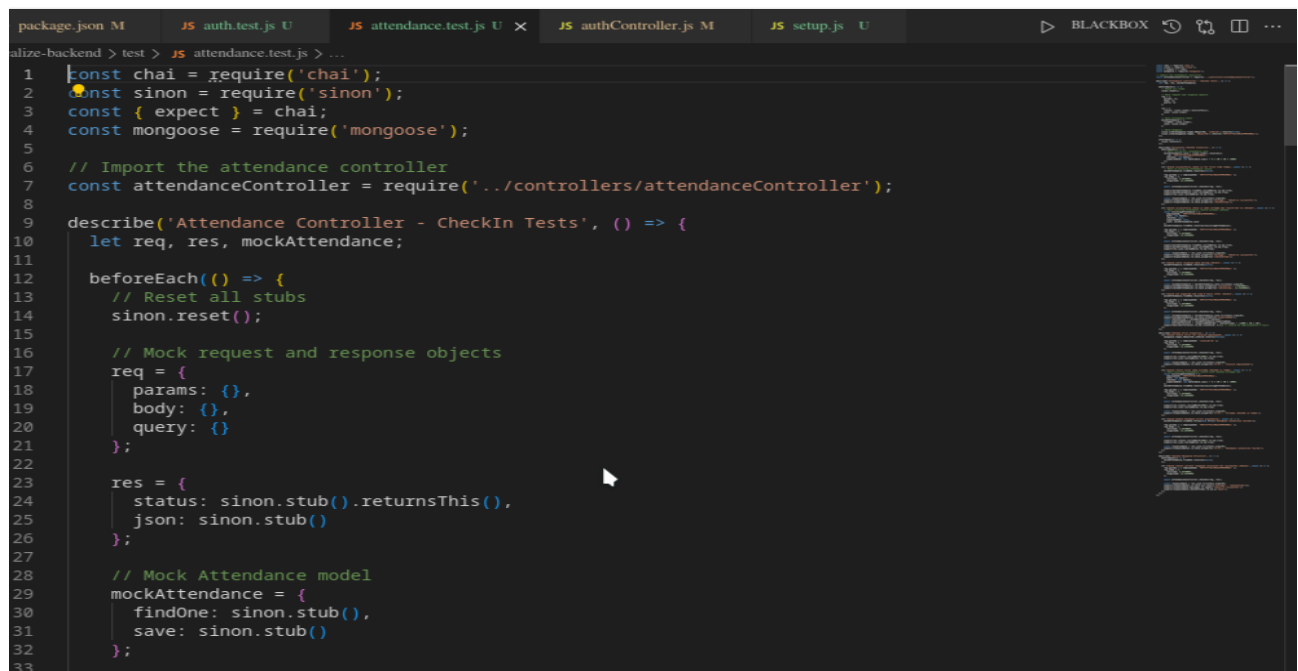
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User check-in: Demonstrates successful user check-in.

```
Successful CheckIn Scenarios
CheckIn: 507f1f77bcf86cd799439011 { latitude: 3.8370057, longitude: 11.5335042 }
    1) should successfully check in for first time today
CheckIn: 507f1f77bcf86cd799439011 { latitude: 3.8370057, longitude: 11.5335042 }
    2) should successfully check in when already has record but no checkIn
CheckIn: 507f1f77bcf86cd799439011 { latitude: 3.8370057, longitude: 11.5335042 }
    3) should store location data during checkin
CheckIn: 507f1f77bcf86cd799439011 { latitude: 3.8370057, longitude: 11.5335042 }
    4) should set expected end time 8 hours after checkin
CheckIn Error Scenarios
    5) should return error for invalid employeeId
CheckIn: 507f1f77bcf86cd799439011 { latitude: 3.8370057, longitude: 11.5335042 }
    6) should return error when already checked in today
```

Figure 38:check-in test code

•User Attendance: Shows successful attendance registration



```
package.json M JS auth.test.js U JS attendance.test.js X JS authController.js M JS setup.js U
alizer-backend > test > JS attendance.test.js > ...
1  const chai = require('chai');
2  const sinon = require('sinon');
3  const { expect } = chai;
4  const mongoose = require('mongoose');
5
6  // Import the attendance controller
7  const attendanceController = require('../controllers/attendanceController');
8
9  describe('Attendance Controller - CheckIn Tests', () => {
10    let req, res, mockAttendance;
11
12    beforeEach(() => {
13      // Reset all stubs
14      sinon.reset();
15
16      // Mock request and response objects
17      req = {
18        params: {},
19        body: {},
20        query: {}
21      };
22
23      res = {
24        status: sinon.stub().returnsThis(),
25        json: sinon.stub()
26      };
27
28      // Mock Attendance model
29      mockAttendance = {
30        findOne: sinon.stub(),
31        save: sinon.stub()
32      };
33    });
```

Figure 39:user attendance test



CONCLUSION

Unit testing played a vital role in ensuring that core functionalities, such as user, login, and user creation, work as expected. By focusing on isolated testing, we ensured that the app components were reliable and bug-free. In the final phase, we will provide an installation and user guide that will explain how to install and use Realize.

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FILE 7: INSTALLATION GUIDE AND USER GUIDE

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PREAMABLE

The aim of the user manual is to furnish our platform users with systematic guidance on installing and utilizing the system.

CONTENT

INTRODUCTION

I. INSTALLATION OF THE APPLICATION

II. SHOWCASE

CONCLUSION

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INTRODUCTION

This is the ultimate stage of our report. In this phase, we will dissect the requirements for our system, the essential installation process, system and feature accessibility in a step-by-step approach to aid first-time users with easy platform setup. Each process will be accompanied by relevant images. Subsequently, we shall present our application by showcasing the various screens.

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INSTALLATION OF THE APPLICATION

1. System requirements

1. For Developers:

- Operating System:** Windows, macOS, or Linux
- Node.js:** Version 14.x or later
- npm or yarn:** Latest version (for package management)
- MongoDB:** Latest version (for database)
- Visual Studio Code** or any other preferred text editor/IDE
- Git** (for cloning the repository)

2. For Users:

- Device:** Android phone/ Laptop
- Operating System:** windows, mac, Linux, android.
- Internet Connection

2. Installation Guide (For Developers)

2.1. Backend Installation

Step 1: Clone the repository

Step 2: Install dependencies



2.2. Frontend Installation (React JSX)

Step 1: Clone the repository

Step 2: Install dependencies

Step 3: Set up environment variables

- Create a .env file in the frontend project and configure the backend API_BASE_URL

Step 4: Run the expo project with Expo

2.3. Installing the web version

Step 1: Clone the repository

Step 2: Install dependencies

Step 3: Set up environment variables

- Create a .env file in the root directory of the folder. Add the following variables:

Step 4: Start the app: Run the following command to start the web server for the app

“npm run dev”

2. Installation Guide (For Users)

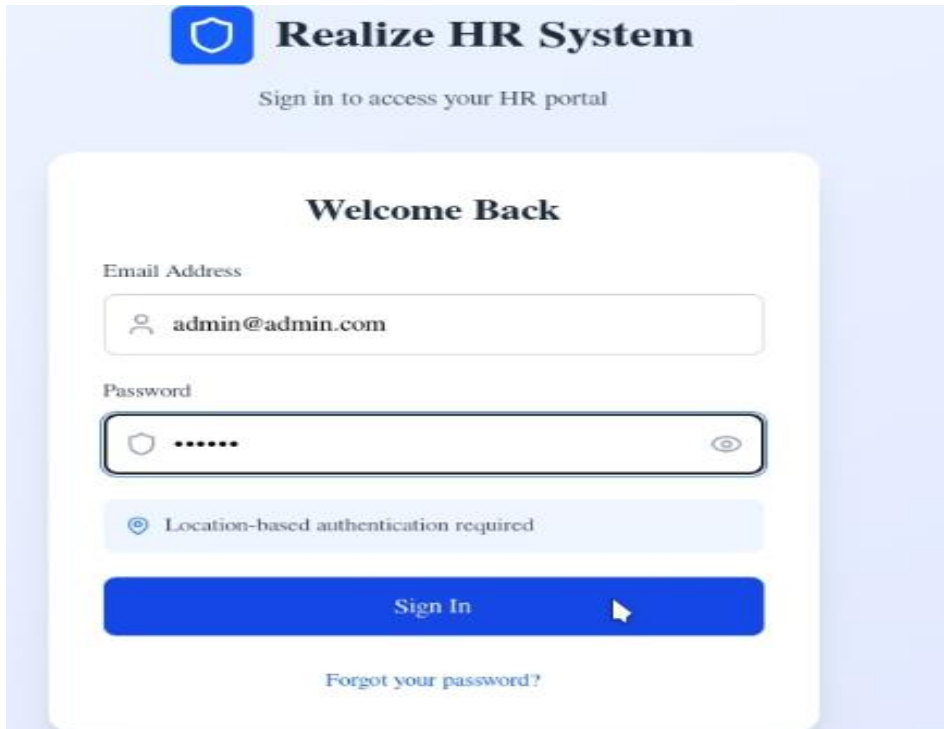
- Visit the realize website then login.



• II. SHOWCASE

1. Login:

- Open the app and enter your registered email and password.
- Click Sign in to access your account.



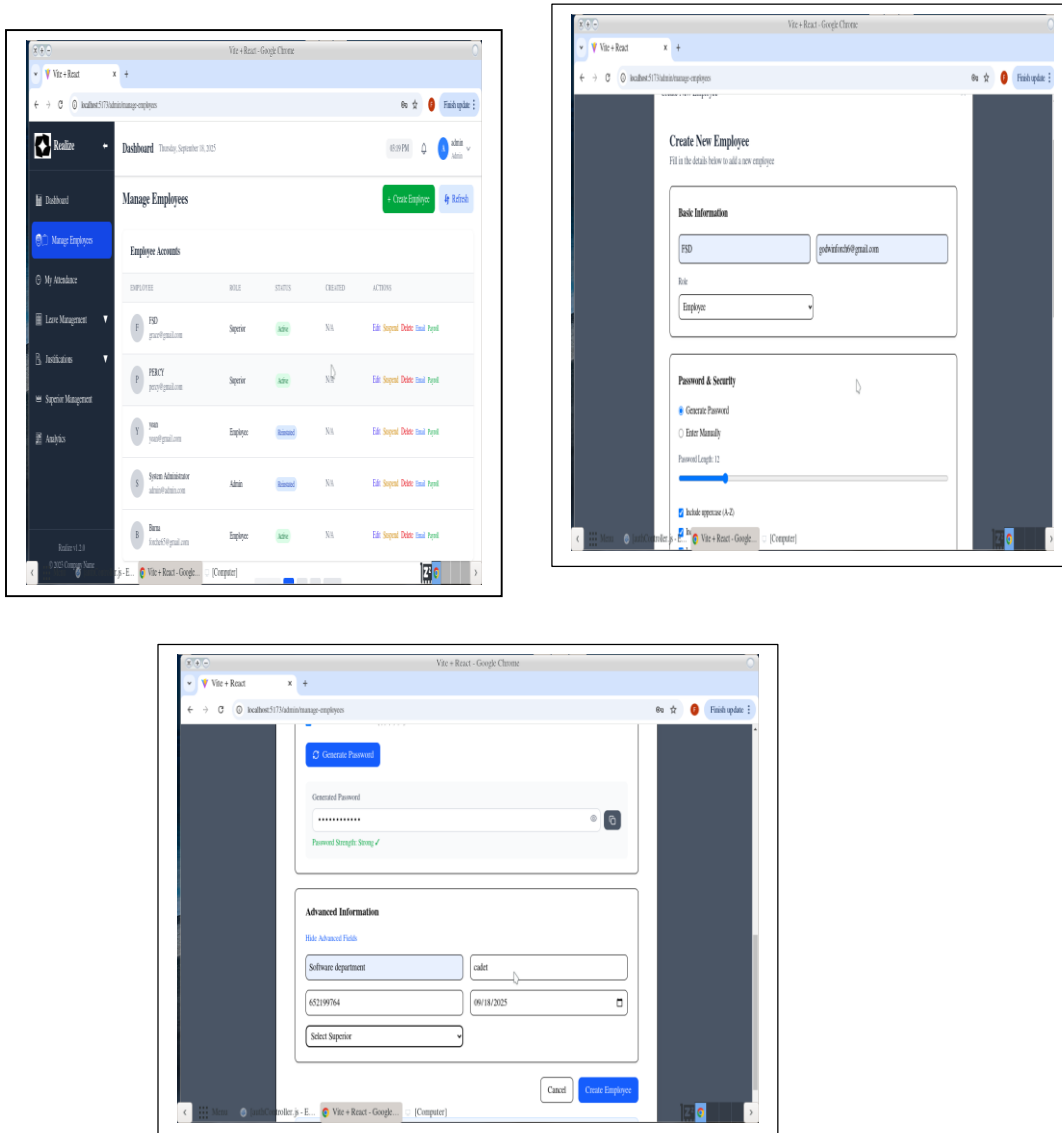
The screenshot shows the login interface of the 'Realize HR System'. At the top, there is a blue shield icon and the text 'Realize HR System'. Below this, it says 'Sign in to access your HR portal'. The main content area is a white card with the heading 'Welcome Back'. It contains two input fields: 'Email Address' with the value 'admin@admin.com' and 'Password' with masked characters. Below the password field, there is a blue button labeled 'Sign In' and a link 'Forgot your password?'. A message 'Location-based authentication required' is displayed in a light blue box above the 'Sign In' button.

Figure 40: login

3. Employee creation

- When you are login navigate to manage employee.
- Click on create employee.

Figure 41: employee creation



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GENERAL CONCLUSION

As our internship at **REALIZE** draws to a close, we reflect on the progress and valuable knowledge gained throughout this enriching experience. This period has allowed us to strengthen our technical expertise, adapt to real-world challenges, and develop essential teamwork and problem-solving skills.

Our project focused on the “**Conception and Realization of an Employee Attendance and Payroll Management Web Application.**” The main objective was to create a reliable system that tracks employee attendance, calculates work hours, manages payroll, and streamlines leave requests and justifications. To achieve this, we began with a thorough analysis of the company’s needs and identified the limitations of the existing manual processes.

Using **UML methodology**, we designed the architecture of the system through diagrams such as class, package, component, and deployment models. For development, we proposed a **modern stack**: a responsive web front-end (React), a secure back-end (Node.js express,), and a non-relational database (Mongo DB) for data integrity. The system also integrates features like location-based login, leave approval workflows, and automated payroll calculations.

The resulting solution provides employees, supervisors, and HR with a **user-friendly, transparent, and efficient platform** that not only improves accountability but also reduces administrative workload. Furthermore, by incorporating security mechanisms and scalability in its design, the system can be adapted to support multiple sites and future extensions.

In conclusion, this project has demonstrated the potential of digital solutions to transform traditional workforce management. The Realize Employee Attendance & Payroll system lays the foundation for greater efficiency, fairness, and reliability in employee management. Moving forward, the system can be enhanced with mobile integration, advanced analytics, and AI-driven insights, ensuring it continues to meet the evolving needs of the organization.

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PERSPECTIVES

Considering the fact that this project can be improved to a higher extend, as perspectives we can bring some amelioration such as:

- Implementing a chat between the employees.
- Adding a QR scan for authentication to ensure more secure authentication.
- Adding a permission take flied in case of any emergency.

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ANNEXES

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BIBLIOGRAPHY

- OMG Unified Modelling Language 2.5. Object Management Group. March 2015. 794 pages.
- Gamma, Erich, et al. Design Patterns: Elements of Reusable Object-Oriented Software. Addison-Wesley, 1994. 395 pages.
- Fowler, Martin. Patterns of Enterprise Application Architecture. Addison-Wesley Professional, 2002.
- Rouse, Margaret. What is MERN Stack? TechTarget. Last modified September 2023.
- Eloquent JavaScript: A Modern Introduction to Programming, Marijn Havebeke 472 pages.

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WEBOGRAPHY

- [https://www.visual-paradigm.com/guide/uml-unified-modeling
language/umldiagrams/](https://www.visual-paradigm.com/guide/uml-unified-modeling-language/umldiagrams/) UML Diagrams Guide.
- [https://developer.mozilla.org/en
US/docs/Learn_web_development/Extensions/Server-
side/Express_Nodejs/Introduction](https://developer.mozilla.org/en-US/docs/Learn_web_development/Extensions/Server-side/Express_Nodejs/Introduction).
- <https://reactnative.dev/> learn react.
- <http://www.w3schools.com/react/> tutorial course.

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